BILINEAR RANK-2 FACTORIZATION OF INTEGRATED WRONSKIAN KERNELS, PURE-SQUARE COMPLETION, AND THE $1+2d>1$ MONTGOMERY–VAUGHAN EXPONENT JUMP IN THE CRITICAL STRIP

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Abstract. We study the off-diagonal cross-terms of the parity-reduced Wronskian $\widetilde{W}_{\mathcal{M}}(d,t)$ and its integral $\widetilde{\mathcal{E}}_{\mathcal{M}}(d,t)$ for Riemann's Ξ-function on $d\in(0,\frac{1}{2}]$. First, every off-diagonal pair $1\le m<n\le N(t)=\lfloor\sqrt{t/2\pi}\rfloor$ obeys $\mathcal{H}^{+}_{mn}(d,t)/\mathcal{D}_{1,1}(d,t)<(mn)^{-d}\le 2^{-d}<1$, and the integrated cross-term kernel factors into the bilinear rank-2 form $w^{\mathrm{int}}_{mn}(d,t)\equiv 2d^{-2}(\cosh(d\omega_m(t))\cosh(d\omega_n(t))-1)$, reducing $\mathcal{O}(N(t)^2)$ double sums to $\mathcal{O}(N(t))$ sums with a phase transition $N(t)^{1-4d}$ at $d=\frac{1}{4}$. Second, adding the critical-line ground state cancels the -1 boundary term pair by pair, completing the $(m,n\ge 2)$ sector into the pure square $+d^{-2}\|\Psi_{\ge 2}(d,t)\|^2\ge 0$ and eliminating the $1+\sqrt{2}$ penalty. Finally, passing from L^1 to L^2 jumps the exponent from $\frac{1}{2}+d<1$ to $1+2d>1$ for all $d>0$, which with Preissmann's Hilbert constant $c_{\mathrm{MV}}=\frac{4}{3}\sqrt{6}$ on windows $\Delta T_{\mathrm{MV}}(d,t)\asymp N(t)^{1-2d}=o(t^{1/2})$ gives $X^{(r)}_{\mathrm{MV},\infty}(d)=\mathcal{O}(1)$ and $\theta^{(r)}_{\mathrm{MV}}(d,t)=\mathcal{O}(1/\ln t)\to 0$.

1. Introduction and Main Results

In the analytic theory of the Riemann zeta function $\zeta(s)$ ($s=\sigma+\mathrm{i}t=\frac{1}{2}+d+\mathrm{i}t$) and Riemann's completed function $\Xi(z):=\xi(\frac{1}{2}+\mathrm{i}z)=R(d,t)+\mathrm{i}\,I(d,t)$ ($z=t-\mathrm{i}d$) [22, 26], the monotonicity of $|\Xi(t-\mathrm{i}d)|^2$ away from the critical line $d=0$ is governed by the transverse Wronskian

$$W(d,t):=\frac{1}{2}\frac{\partial}{\partial d}\big|\Xi(t-\mathrm{i}d)\big|^2=R\,\partial_d R+I\,\partial_d I=R\,\partial_t I-I\,\partial_t R, \tag{1.1}$$

whose strict positivity for $d>0$ is equivalent to the Speiser–Hinkkanen–Lagarias monotonicity criterion $\mathrm{Re}(\xi'(s)/\xi(s))>0$ [25, 6, 9] and closely linked to the Pólya–Jacobi kernel $\Phi(u)$ [8, 20], Turán–Jensen total positivity [3], Wigner phase-space quasi-probabilities [27, 7], and de Branges Hilbert spaces [2]. In Papers I–VII of this series [11, 12, 13, 14, 15, 16, 17], we introduced the parity-reduced Wronskian $\widetilde{W}(d,t):=W(d,t)/d$, the universal mother-wavelet decomposition $\Phi(u)=\sum_{n=1}^{\infty}n^{-1/2}\varphi_1(u+\frac{1}{2}\ln n)$, and finite-divisor sieved multipliers $\mathcal{M}(z)$ preserving critical-line mirror symmetry.

When the mother-wavelet expansion of $\Xi_{\mathcal{M}}(z):=\Xi(z)\mathcal{M}(z)$ is evaluated via complex saddle points up to the Riemann–Siegel horizon $N(t):=\lfloor\sqrt{t/2\pi}\rfloor$ [24, 4], the parity-reduced transverse gradient decomposes into a strictly positive, oscillation-free diagonal ($m=n$) and $\binom{N(t)}{2}$ oscillatory off-diagonal cross-pairs ($1\le m<n\le N(t)$). Controlling these $\binom{N(t)}{2}$ off-diagonal pairs across the critical strip $0<d\le\frac{1}{2}$ as $t\to\infty$ confronts four fundamental analytical questions:

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(1) *Quadratic pair complexity $\mathcal{O}(N(t)^2)$ and hyperbolic off-diagonal scaling.* At height $t = 10^8$ ($N(t) = 3989$), there are $\binom{3989}{2} = 7{,}954{,}066$ active cross-pairs (m, n), each weighted by coupled hyperbolic frequencies $\omega_{mn}^{+}(t) = \frac{1}{2}\ln\frac{t}{2\pi mn}$ and $\omega_{mn}^{-} = \frac{1}{2}\ln\frac{n}{m}$. Can the integrated off-diagonal kernel be factored exactly into separable one-dimensional sums in m and n?

(2) *The $(m, n \geq 2)$ mutual cross-term sign problem and the $1 + \sqrt{2}$ root penalty.* In worst-case triangle-inequality bounds, assigning a simultaneous minus sign to both the $(1, n)$ cross-sum $-S_{1,\mathrm{off}}/u$ and the $(m, n \geq 2)$ mutual cross-sum $-S_{2,\mathrm{off}}/u^2$ ($u = (\ln X)^r$) produces the quadratic equation $1 - 2Y - Y^2 = 0$, inflating the critical sieve threshold by $1 + \sqrt{2} \approx 2.41421356$. Can the $(m, n \geq 2)$ sector actually become negative?

(3) *The Euler–Maclaurin phase transition at $d = \frac{1}{4}$ ($\sigma = \frac{3}{4}$).* How do the integrated off-diagonal sums $S_{1,\mathrm{off}}^{\mathrm{int},(r)}(d, t)$ and $S_{2,\mathrm{off}}^{\mathrm{int},(r)}(d, t)$ scale relative to the ground-state diagonal $\mathcal{D}_{1,1}^{\mathrm{int}}(d, t) \sim \frac{1}{2d^2}N(t)^{2d}$ across the inner strip $d \in (0, \frac{1}{4})$ versus the outer strip $d \in (\frac{1}{4}, \frac{1}{2}]$?

(4) *Overcoming the L^1 divergence barrier $\frac{1}{2} + d < 1$ inside $0 < d < \frac{1}{2}$.* Because individual harmonic amplitudes scale as $|b_n| n^{-(1/2+d)}$, any phase-blind L^1 triangle inequality diverges as $N(t)^{1/2-d}$ for $0 < d < \frac{1}{2}$ [23, 10, 1]. How does incorporating phase oscillation via the Montgomery–Vaughan–Preissmann Hilbert inequality [18, 21] and Gallagher's Sobolev inequality [5] alter the critical exponent across $d > 0$?

In the present paper (Paper VIII), we resolve all four questions analytically and computationally up to $t = 10^8$. Our main results are summarized in the following theorem.

Theorem 1.1 (Bilinear Rank-2 Factorization, Pure-Square Completion, and the $1 + 2d > 1$ Exponent Jump)**.** *Let $s = \frac{1}{2} + d + \mathrm{i}t$ and $z = t - \mathrm{i}d$ with $d > 0$ and $t > 4\pi$, let $L := \ln(t/2\pi)$, $x_t := \sqrt{t/2\pi}$, $N(t) := \lfloor x_t \rfloor$, and $\omega_n(t) := \frac{1}{2}\ln\frac{t}{2\pi n^2} = \frac{1}{2}L - \ln n$ for $1 \leq n \leq N(t)$. Under the smooth order-r Möbius filter $P_r(x) = 1 - (1 - x)^r$ ($r \in \mathbb{N}_{\geq 1}$) with sieved coefficients $b_n^{(r)}(X) = \sum_{k|n,\, k \leq X} \mu(k) P_r(\frac{\ln(X/k)}{\ln X})$, the following four structural laws hold unconditionally:*

(i) ***Universal $(mn)^{-d} \leq 2^{-d}$ Hyperbolic Diagonal Advantage and Order-r Generalized von Mangoldt Scaling.*** *In the parity-reduced Riemann–Siegel mirror gradient, every off-diagonal pair $1 \leq m < n \leq N(t)$ satisfies the exact decomposition $\sinh(2d\omega_{mn}^{+}(t)) \equiv (mn)^{-d}\sinh(2d\omega_1(t)) - \sinh(d\ln(mn))(t/2\pi)^{-d}$ and the universal pointwise inequality*

$$\frac{\mathcal{H}_{mn}^{+}(d, t)}{\mathcal{D}_{1,1}(d, t)} < \left(1 - \frac{\ln(mn)}{\ln(t/2\pi)}\right)(mn)^{-d} < (mn)^{-d} \leq 2^{-d} < 1 \qquad (\forall\, d > 0). \tag{1.2}$$

Moreover, $b_1^{(r)}(X) \equiv 1$ and $b_n^{(r)}(X) = \frac{\Lambda_r(n)}{(\ln X)^r}$ for all $2 \leq n \leq X$, where $\Lambda_r(n) := -\sum_{k|n}\mu(k)(\ln k)^r$ obeys the parity alternation law $|\Lambda_r(n)| \equiv (-1)^{\omega(n)-1}\Lambda_r(n) \geq 0$, vanishes identically whenever $\omega(n) > r$, satisfies the radical bound $|\Lambda_r(n)| \leq (\ln \mathrm{rad}(n))^r \leq (\ln n)^r$, and obeys the Stirling–Dirichlet convolution asymptotics $\sum_{n \leq x}|\Lambda_r(n)| \sim \left(\sum_{m=1}^{r} \frac{1}{(r-1)!}\left[\begin{smallmatrix} r \\ m \end{smallmatrix}\right]\right) x(\ln x)^{r-1} = r\,x\,(\ln x)^{r-1}$.

(ii) ***Exact Bilinear Rank-2 Factorization of Integrated Wronskian Kernels and Phase Transition at $d = \frac{1}{4}$.*** *Under the hyperbolic low-pass filter $\mathcal{F}_{\mathrm{int}}(x) := \frac{\tanh(x/2)}{x/2} \in (0, 1)$ of the integrated sieved energy clamp $\widetilde{\mathcal{E}}_{\mathcal{M}}(d, t) := \frac{1}{d^2}|\Xi_{\mathcal{M}}(t)|^2 + \frac{2}{d^2}\int_0^d u\,\widetilde{W}_{\mathcal{M}}(u, t)\,\mathrm{d}u$, the integrated Gabcke–Olver remainder is damped relative to the leading diagonal by $\cosh^{-2}(\frac{1}{4}dL) < 1$, and the integrated off-diagonal weight factors identically into the bilinear rank-2 form*

$$w_{mn}^{\mathrm{int}}(d, t) \equiv \frac{2\big(\cosh(d\omega_m(t))\cosh(d\omega_n(t)) - 1\big)}{d^2} \qquad (1 \leq m < n \leq N(t)). \tag{1.3}$$

Consequently, the $\mathcal{O}(N(t)^2)$ double sums $S_{1,\mathrm{off}}^{\mathrm{int},(r)}(d, t)$ and $S_{2,\mathrm{off}}^{\mathrm{int},(r)}(d, t)$ reduce identically to the $\mathcal{O}(N(t))$ one-dimensional sums $\Sigma_{\cosh}^{(r)}(d, t) := \sum_{n=2}^{N(t)} \frac{|\Lambda_r(n)|}{\sqrt{n}}\cosh(d\omega_n(t))$ and $\Sigma_0^{(r)}(t) :=$

$\sum_{n=2}^{N(t)} \frac{|\Lambda_r(n)|}{\sqrt{n}}$, *and satisfy for* $d \in (0, \frac{1}{2})$ *as* $N(t) \to \infty$*:*

$$\begin{aligned} \frac{S_{1,\mathrm{off}}^{\mathrm{int},(r)}(d,t)}{\mathcal{D}_{1,1}^{\mathrm{int}}(d,t)} &\sim \frac{8r}{1-4d^2}\, N(t)^{\frac{1}{2}-d} (\ln N(t))^{r-1}, \\ \frac{S_{2,\mathrm{off}}^{\mathrm{int},(r)}(d,t)}{\big(\mathcal{D}_{1,1}^{\mathrm{int}}(d,t)\big)^2} &\sim \frac{256 r^2 d^4 (1-2d^2)}{(1-4d^2)^2}\, N(t)^{1-4d} (\ln N(t))^{2(r-1)}, \end{aligned} \tag{1.4}$$

while the pure cosh*-envelope* $S_{2,\mathrm{tot}}^{\cosh,(r)}(d,t) := \frac{2}{d^2}(\Sigma_{\cosh}^{(r)}(d,t))^2$ *satisfies* $S_{2,\mathrm{tot}}^{\cosh,(r)}/(\mathcal{D}_{1,1}^{\mathrm{int}})^2 \sim \frac{32 r^2 d^2}{(1-4d^2)^2} N(t)^{1-4d} (\ln N(t))^{2(r-1)}$*, exhibiting an exact phase transition at* $d = \frac{1}{4}$ *(*$\sigma = \frac{3}{4}$*), above which (*$d \in (\frac{1}{4}, \frac{1}{2}]$*) the quadratic ratio decays polynomially to* 0 *as* $N(t)^{-(4d-1)}$*.*

(iii) ***Pure-Square Completion, Pairwise*** -1 ***Boundary Cancellation, and Identical Elimination of*** $1+\sqrt{2}$***.*** *When the critical-line ground-state energy* $\mathcal{G}_0(d,t) = \frac{1}{d^2}|\Xi_{\mathcal{M},\mathrm{main}}(t)|^2$ *is combined with the Wronskian integral* $\mathcal{I}_{\mathrm{int}}(d,t)$*, the* -1 *boundary term at* $u = 0$ *in* (1.3) *cancels identically pair by pair against* $\frac{2}{d^2}\cos\phi_m \cos\phi_n$*. The entire* $(m, n \geq 2)$ *sector forms a non-negative pure square with a plus sign (*$\mathcal{E}_{\geq 2}(d,t) \equiv +\frac{1}{d^2}\|\Psi_{\geq 2}(d,t)\|^2 \geq 0$*), yielding*

$$\widetilde{\mathcal{E}}_{\mathcal{M},\mathrm{main}}(d,t) \equiv \frac{1}{d^2}\big\|\psi_1(d,t) + \Psi_{\geq 2}(d,t)\big\|^2 \geq \frac{1}{d^2}\Big(\max\big(0,\ \|\psi_1(d,t)\| - \|\Psi_{\geq 2}(d,t)\|\big)\Big)^2, \tag{1.5}$$

replacing $1-2Y-Y^2 = 0$ *by the completed square* $(1-Y)^2 = 1-2Y+Y^2 \geq 0$*, and eliminating the root penalty* $1+\sqrt{2}$ *identically (*$\mathcal{Q}_r^{\cosh,\mathrm{sq}}(d,t) \equiv (1+\sqrt{2})^{-1}\mathcal{Q}_r^{\cosh,\mathrm{old}}(d,t)$*).*

(iv) ***The*** $1+2d > 1$ ***Exponent Jump in*** L^2 ***and Preissmann–Montgomery–Vaughan Hilbert Bounds.*** *For every* $d > 0$ *across the critical strip, squaring the harmonic amplitudes produces an exponent jump from* $\frac{1}{2}+d < 1$ *to* $1+2d > 1$*, so the forward* L^2 *diagonal converges uniformly (*$\mathcal{E}_{\mathrm{diag}}^{\mathrm{fwd}}(d,t) < (\ln X)^{-2r}\mathcal{S}_{\Lambda_r^2}(d) < \infty$*) while the reflected diagonal decays polynomially to zero (*$\mathcal{E}_{\mathrm{diag}}^{\mathrm{ref}}(d,t) < \frac{\max b_n^2}{2d} N(t)^{-2d} \to 0$*). Combined with Preissmann's sharp Hilbert constant* $c_{\mathrm{MV}} = \frac{4}{3}\sqrt{6}$ *on sub-Riemann–Siegel windows* $\Delta T_{\mathrm{MV}}(d,t) \asymp N(t)^{1-2d} = o(t^{1/2})$*, the critical diagonal* L^2 *sieve length* $X_{\mathrm{MV},\infty}^{(r)}(d) = \exp([\mathcal{S}_{\Lambda_r^2}(d)]^{1/2r}) = \mathcal{O}(1)$ *is finite and* t*-independent, forcing*

$$\theta_{\mathrm{MV}}^{(r)}(d,t) := \frac{\ln X_{\mathrm{MV}}^{(r)}(d,t)}{\ln(t/2\pi)} \sim \frac{[\mathcal{S}_{\Lambda_r^2}(d)]^{1/2r}}{\ln(t/2\pi)} = \mathcal{O}\Big(\frac{1}{d^{1+1/2r}\ln t}\Big) \longrightarrow 0 \qquad (t \to \infty). \tag{1.6}$$

The canonical quantitative constants governing Theorem 1.1 are given explicitly by

$$\begin{aligned} 1+\sqrt{2} &= 2.4142135623\ldots, & c_{\mathrm{MV}} &= \frac{4}{3}\sqrt{6} = 3.2659863237\cdots < \pi\sqrt{2}, \\ X_{\mathrm{krit}}^{\mathrm{sq},(3)}(0.10, 10^4) &= 9.99246971\cdots < 10, & \theta_{\mathrm{sq}}^{(3)}(0.10, 10^4) &= 0.31222022\cdots < \frac{1}{3}, \\ \binom{N(10^8)}{2} &= \binom{3989}{2} = 7{,}954{,}066, & \theta_{\mathrm{MV}}^{(3)}(0.35, 10^8) &= 0.18889201\cdots < \frac{1}{5}. \end{aligned} \tag{1.7}$$

1.1. **Proof overview.** The proof of Theorem 1.1 rests on three interconnected analytical ingredients:

(1) *Hyperbolic difference decomposition and bilinear rank-2 kernel factorization (Sections 2 and 3).* Writing $2d\omega_{mn}^{+}(t) = d\ln(t/2\pi) - d\ln(mn)$, the identity $\sinh(A-B) = \mathrm{e}^{-B}\sinh(A) - \sinh(B)\mathrm{e}^{-A}$ immediately isolates the pointwise damping factor $(mn)^{-d} \leq 2^{-d} < 1$ for every off-diagonal pair $1 \leq m < n \leq N(t)$. Under transverse integration $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$, each hyperbolic

mode $\mathrm{sinc}_{\mathrm{hyp}}(2d\omega)$ is multiplied by the low-pass filter $\mathcal{F}_{\mathrm{int}}(2d\omega) = \tanh(d\omega)/(d\omega)$, converting $2\omega^2\mathcal{F}_{\mathrm{int}}(2d\omega)\mathrm{sinc}_{\mathrm{hyp}}(2d\omega)$ into $(\cosh(2d\omega)-1)/d^2$. Applying the hyperbolic addition law $\cosh(d(\omega_m+\omega_n))+\cosh(d(\omega_m-\omega_n)) \equiv 2\cosh(d\omega_m)\cosh(d\omega_n)$ to $\omega_{mn}^{\pm} = \frac{1}{2}(\omega_m \pm \omega_n)$ yields the exact rank-2 factorization (1.3), collapsing $\mathcal{O}(N(t)^2)$ pair sums to $\mathcal{O}(N(t))$ one-dimensional sums and establishing the Euler–Maclaurin phase transition (1.4) at $d = \frac{1}{4}$ via the Stirling–Dirichlet asymptotics of $|\Lambda_r(n)|$.

(2) *Pairwise -1 boundary cancellation and Euclidean pure-square completion (Section 4).* Expanding the Euclidean inner product $\frac{1}{d^2}\langle\psi_m(d,t),\psi_n(d,t)\rangle$ of the two-dimensional harmonic vectors $\psi_n(d,t) = \frac{b_n}{\sqrt{n}}(\sqrt{2}\cosh(d\omega_n)\cos\phi_n, \sqrt{2}\sinh(d\omega_n)\sin\phi_n)$ and using $2\cos\phi_m\cos\phi_n = \cos(\phi_m-\phi_n)+\cos(\phi_m+\phi_n)$, we prove that the critical-line ground-state contribution at $u=0$ cancels the -1 boundary term in (1.3) identically for every pair (m,n). Consequently, the higher-shell $(m,n\geq 2)$ sector equals $+\frac{1}{d^2}\|\Psi_{\geq 2}(d,t)\|^2 \geq 0$, completing $(1-Y)^2 = 1-2Y+Y^2 \geq 0$ and removing the factor $1+\sqrt{2}$ identically.

(3) *The $1+2d>1$ exponent jump in L^2 and Preissmann–Montgomery–Vaughan / Gallagher–Sobolev transfer (Section 5).* Squaring the forward harmonic amplitude $|b_n|n^{-(1/2+d)}$ shifts the Dirichlet exponent from $\frac{1}{2}+d<1$ to $1+2d>1$ for every $d>0$, rendering the infinite diagonal series $\mathcal{S}_{\Lambda_r^2}(d) = \sum_{n=2}^{\infty}\Lambda_r(n)^2 n^{-(1+2d)} < \infty$ convergent while the gamma-ratio bound $\rho_\Gamma(d,t)^2 < N(t)^{-4d}$ forces the reflected diagonal $\mathcal{E}_{\mathrm{diag}}^{\mathrm{ref}}(d,t) < \frac{\max b_n^2}{2d}N(t)^{-2d} \to 0$. Applying the generalized Hilbert inequality with Preissmann's constant $c_{\mathrm{MV}} = \frac{4}{3}\sqrt{6}$ on sub-Riemann–Siegel windows $\Delta T_{\mathrm{MV}}(d,t) \asymp N(t)^{1-2d} = o(t^{1/2})$ and Gallagher's Sobolev inequality bounds the mean-square and pointwise cross-terms by $\mathcal{O}(1)$ diagonal series.

Together, ingredients (1), (2), and (3) establish Theorem 1.1.

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma+\mathrm{i}t = \frac{1}{2}+d+\mathrm{i}t$ with transverse distance $d = \sigma-\frac{1}{2}\geq 0$ and complex coordinate $z = -\mathrm{i}(s-\frac{1}{2}) = t-\mathrm{i}d$. We write $L := \ln(t/2\pi)$, $x_t := \sqrt{t/2\pi} = \mathrm{e}^{L/2}$, and $N(t) := \lfloor x_t\rfloor$ for the Riemann–Siegel saddle-point horizon. For each integer $1\leq n\leq N(t)$, the local logarithmic frequency is $\omega_n(t) := \frac{1}{2}\ln\frac{t}{2\pi n^2} = \frac{1}{2}L-\ln n = \ln(x_t/n)\geq 0$, and for $1\leq m<n\leq N(t)$ we set

$$\begin{aligned} \omega_{mn}^{+}(t) &:= \frac{1}{2}\ln\Big(\frac{t}{2\pi mn}\Big) = \frac{\omega_m(t)+\omega_n(t)}{2}, \quad \Omega_{mn}^{+}(t) := 2\omega_{mn}^{+}(t) = \omega_m(t)+\omega_n(t), \\ \omega_{mn}^{-} &:= \frac{1}{2}\ln\Big(\frac{n}{m}\Big) = \frac{\omega_m(t)-\omega_n(t)}{2}, \qquad\qquad \Omega_{mn}^{-} := 2\omega_{mn}^{-} = \omega_m(t)-\omega_n(t). \end{aligned} \tag{1.8}$$

The hyperbolic sinc function is $\mathrm{sinc}_{\mathrm{hyp}}(x) := \frac{\sinh x}{x}$ for $x\neq 0$ with $\mathrm{sinc}_{\mathrm{hyp}}(0) := 1$, and the hyperbolic low-pass filter is $\mathcal{F}_{\mathrm{int}}(x) := \frac{\tanh(x/2)}{x/2}$ for $x\neq 0$ with $\mathcal{F}_{\mathrm{int}}(0) := 1$. For an integer $n\geq 1$, $\mu(n)$ denotes the Möbius function, $\omega(n) := \sum_{p\mid n}1$ the number of distinct prime divisors of n, and $\mathrm{rad}(n) := \prod_{p\mid n}p$ the squarefree radical of n. Unsigned Stirling numbers of the first kind are denoted by $\left[\begin{smallmatrix} r \\ m \end{smallmatrix}\right]$. Implied constants in Vinogradov ($\ll$) or Landau ($\mathcal{O}$, $\asymp$) notation are absolute unless subscripts indicate dependence on d or r.

2. The $(mn)^{-d}\leq 2^{-d}$ Hyperbolic Diagonal Advantage and Order-r Generalized von Mangoldt Scaling

In the symmetrized Riemann–Siegel mirror representation of $\Xi_{\mathcal{M}}(t-\mathrm{i}d)$ established in Papers V and VI [15, 16], factoring out the archimedean gamma envelope yields the main harmonic sum of length $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$, whose parity-reduced transverse derivative $\frac{1}{2d}\partial_d\widetilde{\mathcal{Z}}_N(d,t) = \mathcal{D}_N(d,t)+$

$\mathcal{C}_N(d,t)$ splits into the strictly positive diagonal sum

$$\mathcal{D}_N(d,t) := \frac{1}{d}\sum_{n=1}^{N(t)} \frac{b_n^2}{n}\,\omega_n(t)\sinh\bigl(2d\,\omega_n(t)\bigr) > 0 \tag{2.1}$$

and the oscillatory off-diagonal cross-sum

$$\begin{aligned}\mathcal{C}_N(d,t) := \frac{2}{d}\sum_{1\le m<n\le N(t)} \frac{b_m b_n}{\sqrt{mn}}\Bigl[&\omega^+_{mn}(t)\sinh\bigl(2d\,\omega^+_{mn}(t)\bigr)\cos\Bigl(t\ln\frac{n}{m}\Bigr)\\ &+\omega^-_{mn}\sinh\bigl(2d\,\omega^-_{mn}\bigr)\cos\bigl(2\vartheta(t)-t\ln(mn)\bigr)\Bigr].\end{aligned} \tag{2.2}$$

We first prove that every off-diagonal pair $1 \le m < n \le N(t)$ suffers an exact pointwise hyperbolic damping factor $(mn)^{-d} \le 2^{-d} < 1$ relative to the ground-state diagonal $\mathcal{D}_{1,1}(d,t) := \frac{1}{d}\omega_1(t)\sinh(2d\omega_1(t))$.

Theorem 2.1 (Universal $(mn)^{-d} \le 2^{-d}$ Hyperbolic Diagonal Advantage)**.** *Let $d > 0$, $t > 4\pi$, and $1 \le m < n \le N(t) = \lfloor\sqrt{t/2\pi}\rfloor$. Define the primary hyperbolic cross-weight $\mathcal{H}^+_{mn}(d,t) := \frac{1}{d}\omega^+_{mn}(t)\sinh(2d\omega^+_{mn}(t))$ and secondary cross-weight $\mathcal{H}^-_{mn}(d) := \frac{1}{d}\omega^-_{mn}\sinh(2d\omega^-_{mn})$. Then the following exact hyperbolic decomposition holds:*

$$\sinh\bigl(2d\,\omega^+_{mn}(t)\bigr) \equiv (mn)^{-d}\sinh\bigl(2d\,\omega_1(t)\bigr) - \sinh\bigl(d\ln(mn)\bigr)\left(\frac{t}{2\pi}\right)^{-d}, \tag{2.3}$$

and consequently every off-diagonal pair (m,n) with $1 \le m < n \le N(t)$ satisfies the strict pointwise inequalities

$$\frac{\mathcal{H}^+_{mn}(d,t)}{\mathcal{D}_{1,1}(d,t)} < \left(1 - \frac{\ln(mn)}{\ln(t/2\pi)}\right)(mn)^{-d} < (mn)^{-d} \le 2^{-d} < 1, \tag{2.4}$$

while $\mathcal{H}^-_{mn}(d)/\mathcal{D}_{1,1}(d,t) \le \frac{1}{2}(t/2\pi)^{-d/2}(1-(t/2\pi)^{-2d})^{-1} = \mathcal{O}((t/2\pi)^{-d/2})$.

Proof. Set $A := 2d\omega_1(t) = d\ln(t/2\pi) > 0$ and $B := d\ln(mn) > 0$, since $1 \le m < n$ implies $mn \ge 2 > 1$. Because $2d\omega^+_{mn}(t) = A - B \ge 0$, expanding the hyperbolic sine gives

$$\sinh(A-B) = \mathrm{e}^{-B}\,\frac{\mathrm{e}^{A}-\mathrm{e}^{-A}}{2} - \frac{\mathrm{e}^{B}-\mathrm{e}^{-B}}{2}\,\mathrm{e}^{-A} = (mn)^{-d}\sinh(A) - \sinh(B)\left(\frac{t}{2\pi}\right)^{-d},$$

which proves (2.3). Since $B = d\ln(mn) > 0$, we have $\sinh(B)(t/2\pi)^{-d} > 0$, so $\sinh(2d\omega^+_{mn}(t)) < (mn)^{-d}\sinh(2d\omega_1(t))$. Multiplying by the frequency ratio $\frac{\omega^+_{mn}(t)}{\omega_1(t)} = 1 - \frac{\ln(mn)}{\ln(t/2\pi)} \in (0,1)$ and noting that $mn \ge 2 \implies (mn)^{-d} \le 2^{-d} < 1$ for all $d > 0$ yields (2.4). Finally, for the secondary weight, $1 \le m < n \le N(t) \le \sqrt{t/2\pi}$ implies $0 < \omega^-_{mn} = \frac{1}{2}\ln(n/m) \le \frac{1}{2}\ln N(t) \le \frac{1}{2}\omega_1(t)$, so $\sinh(2d\omega^-_{mn}) \le \sinh(d\omega_1(t)) = \frac{\sinh(2d\omega_1(t))}{2\cosh(d\omega_1(t))} < \frac{1}{2}(t/2\pi)^{-d/2}\sinh(2d\omega_1(t))$. □

Next, we establish the exact arithmetic properties of the generalized von Mangoldt function $\Lambda_r(n)$ generated by the smooth order-r Möbius mollifier $M^{(r)}_X(s) = \sum_{k\le X}\mu(k)P_r(\frac{\ln(X/k)}{\ln X})k^{-s}$ with $P_r(x) = 1-(1-x)^r$.

Proposition 2.2 (Order-r Generalized von Mangoldt Scaling, Parity Law, Stirling–Dirichlet Convolution, and Radical Bound)**.** *Let $r \in \mathbb{N}_{\ge 1}$ and $X > 1$, and let $b^{(r)}_n(X) := \sum_{k\mid n,\,k\le X}\mu(k)P_r(\frac{\ln(X/k)}{\ln X})$ be the Dirichlet coefficients of $\zeta(s)M^{(r)}_X(s)$ with $P_r(x) = 1-(1-x)^r$. Define the generalized von Mangoldt function of order r by $\Lambda_r(n) := -\sum_{k\mid n}\mu(k)(\ln k)^r$. Then:*

(i) ***Ground-state preservation and exact $(\ln X)^{-r}$ scaling on $2 \le n \le X$.*** *We have $b^{(r)}_1(X) = 1$ for all $X > 1$, and for every integer $2 \le n \le X$:*

$$b^{(r)}_n(X) = \frac{\Lambda_r(n)}{(\ln X)^r}. \tag{2.5}$$

In particular, on prime powers $n = p^a \le X$ $(a \ge 1)$, $\Lambda_r(p^a) = (\ln p)^r$ and $b_{p^a}^{(r)}(X) = (\frac{\ln p}{\ln X})^r$.

(ii) ***Exact vanishing on $\omega(n) > r$, parity alternation, and radical bound.*** *For any integer $n = p_1^{a_1} \cdots p_m^{a_m} \ge 2$ with $\omega(n) = m$ distinct prime factors and radical $\operatorname{rad}(n) = p_1 \cdots p_m$, we have $\Lambda_r(n) = 0$ whenever $m > r$, while for $1 \le m \le r$:*

$$|\Lambda_r(n)| \equiv (-1)^{\omega(n)-1}\Lambda_r(n) = \sum_{\substack{j_1+\cdots+j_m=r \\ j_1,\dots,j_m \ge 1}} \frac{r!}{j_1! \cdots j_m!} (\ln p_1)^{j_1} \cdots (\ln p_m)^{j_m} > 0, \tag{2.6}$$

and for all $n \ge 2$ and all $r \ge 1$, the strict universal bound holds:

$$0 \le |\Lambda_r(n)| \le \big(\ln \operatorname{rad}(n)\big)^r \le (\ln n)^r, \qquad \lim_{r\to\infty} |\Lambda_r(n)|^{1/r} = \ln \operatorname{rad}(n) \le \ln n. \tag{2.7}$$

(iii) ***Stirling–Dirichlet convolution asymptotics.*** *As $x \to \infty$, the contribution of integers with $\omega(n) = m$ $(1 \le m \le r)$ to $\Psi_r^{\mathrm{abs}}(x) := \sum_{n\le x} |\Lambda_r(n)|$ is governed by the unsigned Stirling numbers of the first kind $\left[\begin{smallmatrix} r \\ m \end{smallmatrix}\right]$:*

$$\begin{aligned} \Psi_{r,m}^{\mathrm{abs}}(x) &:= \sum_{\substack{n\le x \\ \omega(n)=m}} |\Lambda_r(n)| \sim c_{r,m}\, x\, (\ln x)^{r-1}, \qquad c_{r,m} = \frac{1}{(r-1)!} \begin{bmatrix} r \\ m \end{bmatrix}, \\ \Psi_r^{\mathrm{abs}}(x) &\sim \left(\sum_{m=1}^{r} c_{r,m} \right) x\, (\ln x)^{r-1} = r\, x\, (\ln x)^{r-1}. \end{aligned} \tag{2.8}$$

Proof. (i) For $n = 1$, $b_1^{(r)}(X) = \mu(1)P_r(1) = 1$. For $2 \le n \le X$, every divisor $k \mid n$ satisfies $1 \le k \le n \le X$ and $1 - \frac{\ln(X/k)}{\ln X} = \frac{\ln k}{\ln X}$, so $P_r(\frac{\ln(X/k)}{\ln X}) = 1 - (\frac{\ln k}{\ln X})^r$. Since $\sum_{k|n} \mu(k) = 0$ for $n \ge 2$, summing over $k \mid n$ gives (2.5).

(ii) Because $\mu(k) = 0$ unless k is squarefree, $\Lambda_r(n) = -\sum_{k|\operatorname{rad}(n)} \mu(k)(\ln k)^r$ depends only on $\operatorname{rad}(n) = p_1 \cdots p_m$. By the multinomial theorem, for each subset $S \subseteq \{1, \dots, m\}$, $(\sum_{i\in S} \ln p_i)^r = \sum_{j_1+\cdots+j_m=r,\, \operatorname{supp}(\mathbf{j})\subseteq S} \frac{r!}{j_1!\cdots j_m!} (\ln p_1)^{j_1} \cdots (\ln p_m)^{j_m}$. By inclusion–exclusion over $S \subseteq \{1, \dots, m\}$ with weight $-\mu(\prod_{i\in S} p_i) = (-1)^{|S|-1}$, all monomials with any $j_i = 0$ cancel identically, leaving only strictly positive indices $j_1, \dots, j_m \ge 1$ with overall sign $(-1)^{m-1}$. If $m > r$, no tuple of m positive integers can sum to r, so $\Lambda_r(n) = 0$. If $1 \le m \le r$, every retained multinomial term is positive, proving (2.6). Comparing (2.6) against the full multinomial expansion of $(\ln \operatorname{rad}(n))^r = (\ln p_1 + \cdots + \ln p_m)^r$ (which includes all non-negative tuples $j_i \ge 0$) immediately gives $|\Lambda_r(n)| \le (\ln \operatorname{rad}(n))^r \le (\ln n)^r$, with equality for $m = 1$ ($\operatorname{rad}(p^a) = p$).

(iii) By the Prime Number Theorem in the form $\sum_{p^a \le x} (\ln p)^j \sim x(\ln x)^{j-1}$ $(j \ge 1)$, repeated Stieltjes–Dirichlet convolution of m prime-power factors yields $\sum_{p_1^{a_1}\cdots p_m^{a_m} \le x} (\ln p_1)^{j_1} \cdots (\ln p_m)^{j_m} \sim \frac{(j_1-1)!\cdots(j_m-1)!}{(r-1)!}\, x(\ln x)^{r-1}$ via the beta-integral identity $\int_{\Delta_m} u_1^{j_1-1} \cdots u_m^{j_m-1}\, \mathrm{d}u = \frac{(j_1-1)!\cdots(j_m-1)!}{(r-1)!}$. Dividing by $m!$ to account for unordered prime factors and multiplying by $\frac{r!}{j_1!\cdots j_m!}$ gives

$$c_{r,m} = \frac{r}{m!} \sum_{\substack{j_1+\cdots+j_m=r \\ j_1,\dots,j_m \ge 1}} \frac{1}{j_1 \cdots j_m} = \frac{1}{(r-1)!} \begin{bmatrix} r \\ m \end{bmatrix},$$

since the exponential generating function of unsigned Stirling numbers of the first kind is $\frac{1}{m!}(-\ln(1-z))^m = \sum_{r=m}^{\infty} \left[\begin{smallmatrix} r \\ m \end{smallmatrix}\right] \frac{z^r}{r!}$. Summing over $1 \le m \le r$ yields $\sum_{m=1}^{r} c_{r,m} = \frac{1}{(r-1)!} \sum_{m=1}^{r} \left[\begin{smallmatrix} r \\ m \end{smallmatrix}\right] = \frac{r!}{(r-1)!} = r$. □

TABLE 1. Verification of the hyperbolic diagonal advantage $\mathcal{H}^{+}_{mn}(d,t)/\mathcal{D}_{1,1}(d,t) < (1-\frac{\ln mn}{L})(mn)^{-d} < (mn)^{-d} \le 2^{-d}$ (Theorem 2.1) and the Stirling–Dirichlet decomposition $\sum_{m=1}^{r} c_{r,m} = r$ (Proposition 2.2), computed at 50-digit working precision in `verify_paper8_bilinear_rank2_wronskian.py`.

Height t	Pair (m,n)	d	Exact $\mathcal{H}^{+}_{mn}/\mathcal{D}_{1,1}$	Bound $(1-\frac{\ln mn}{L})(mn)^{-d}$	$(mn)^{-d}$	Identity Residual
100.0	$(1,2)$	0.20	0.5495454014818891	0.6524964423884565	0.8705506	1.336×10^{-51}
100.0	$(1,3)$	0.20	0.3521426322850298	0.4840540551369106	0.8027416	1.336×10^{-51}
1000.0	$(1,2)$	0.25	0.7000435382003077	0.7259301468469843	0.8408964	2.673×10^{-51}
1000.0	$(2,3)$	0.25	0.3615787031013755	0.4131324813220063	0.6389431	1.336×10^{-51}
10000.0	$(1,2)$	0.30	0.7312792489771931	0.7358857282424366	0.8122524	1.069×10^{-50}
10000.0	$(3,5)$	0.30	0.2668776030107682	0.2807740777778522	0.4437850	2.673×10^{-51}
10000.0	$(10,15)$	0.50	0.0237067701103928	0.0261571257282344	0.0816497	0.0

3. Exact Bilinear Rank-2 Factorization of Integrated Wronskian Kernels and the $\mathcal{O}(N(t)^2) \to \mathcal{O}(N(t))$ Reduction

In Paper VII of this series [17], we showed via the fundamental theorem of calculus on $[0,d]$ that for any holomorphic multiplier $\mathcal{M}(z)$ satisfying critical-line reality $\mathcal{M}(\bar{z}) = \overline{\mathcal{M}(z)}$, the integrated sieved energy clamp

$$\widetilde{\mathcal{E}}_{\mathcal{M}}(d,t) := \frac{\left|\Xi_{\mathcal{M}}(t-\mathrm{i}d)\right|^2}{d^2} \equiv \frac{\left(\Xi_{\mathcal{M}}(t)\right)^2}{d^2} + \frac{2}{d^2}\int_0^d u\,\widetilde{W}_{\mathcal{M}}(u,t)\,\mathrm{d}u \tag{3.1}$$

provides a sufficient and necessary condition for zero exclusion off the critical line ($\widetilde{\mathcal{E}}_{\mathcal{M}}(d_0,t_0) > 0 \implies \Xi(t_0-\mathrm{i}d_0)\neq 0$). Because the finite sum on $u\in[0,d]$ consists of smooth hyperbolic functions, term-by-term integration under $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$ is justified by uniform convergence on compact intervals. We first record how this integral operator acts on the Gabcke–Olver remainder bound and on individual hyperbolic modes.

Theorem 3.1 (Hyperbolic Low-Pass Filter $\mathcal{F}_{\mathrm{int}}(x)$ and $\cosh^{-2}(\frac{1}{4}dL)$ Remainder Damping)**.** *For any frequency $\alpha>0$ and transverse distance $d>0$, the integral operator $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$ maps the parity-reduced hyperbolic kernel $\mathrm{sinc}_{\mathrm{hyp}}(u\alpha)=\frac{\sinh(u\alpha)}{u\alpha}$ to*

$$\frac{2}{d^2}\int_0^d u\,\frac{\sinh(u\alpha)}{u\alpha}\,\mathrm{d}u = \frac{2(\cosh(d\alpha)-1)}{d^2\alpha^2} = \left(\frac{\sinh(d\alpha/2)}{d\alpha/2}\right)^2 = \mathcal{F}_{\mathrm{int}}(d\alpha)\,\frac{\sinh(d\alpha)}{d\alpha} < \frac{\sinh(d\alpha)}{d\alpha}, \tag{3.2}$$

where $\mathcal{F}_{\mathrm{int}}(x)=\frac{\tanh(x/2)}{x/2}\in(0,1)$ is strictly decreasing on $(0,\infty)$. In particular, applying $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$ to the pointwise Gabcke–Olver saddle-point remainder bound $B^{(C_0)}_{\widetilde{W}}(u,t) = 0.650\,L\,\mathrm{sinc}_{\mathrm{hyp}}(\frac{1}{2}uL)\,t^{-3/4}\mathrm{e}^{-\pi t/2}$ from Paper V [15, 4, 19] yields $B^{(C_0)}_{\widetilde{\mathcal{E}}}(d,t) = \mathcal{F}_{\mathrm{int}}(\frac{1}{2}dL)B^{(C_0)}_{\widetilde{W}}(d,t) < B^{(C_0)}_{\widetilde{W}}(d,t)$, and reduces the relative remainder ratio against the integrated $(1,1)$ diagonal growth factor $\mathrm{sinc}^2_{\mathrm{hyp}}(\frac{1}{2}dL)$ by

$$\frac{\mathrm{sinc}^2_{\mathrm{hyp}}(\frac{1}{4}dL)}{\mathrm{sinc}^2_{\mathrm{hyp}}(\frac{1}{2}dL)} \equiv \frac{1}{\cosh^2(\frac{1}{4}dL)} < 1 \qquad (\forall\, dL>0). \tag{3.3}$$

Proof. Since $2u\,\frac{\sinh(u\alpha)}{u\alpha} = \frac{2}{\alpha}\sinh(u\alpha)$, direct integration over $u\in[0,d]$ gives $\frac{2(\cosh(d\alpha)-1)}{d^2\alpha^2} = \frac{4\sinh^2(d\alpha/2)}{d^2\alpha^2} = \mathrm{sinc}^2_{\mathrm{hyp}}(d\alpha/2)$. Writing $\sinh(d\alpha) = 2\sinh(d\alpha/2)\cosh(d\alpha/2)$ yields $\mathrm{sinc}^2_{\mathrm{hyp}}(d\alpha/2)/\mathrm{sinc}_{\mathrm{hyp}}(d\alpha) = \frac{\tanh(d\alpha/2)}{d\alpha/2} = \mathcal{F}_{\mathrm{int}}(d\alpha) < 1$. Setting $\alpha=\frac{1}{2}L$ in the numerator

of (3.3) and $\alpha = L$ in the denominator gives $\frac{\operatorname{sinc}^2_{\mathrm{hyp}}(dL/4)}{\operatorname{sinc}^2_{\mathrm{hyp}}(dL/2)} = \big(\frac{\sinh(dL/4)}{dL/4}\big)^2\big(\frac{dL/2}{2\sinh(dL/4)\cosh(dL/4)}\big)^2 = \cosh^{-2}(\frac{1}{4}dL) < 1$. □

We now prove the central structural identity of this paper: under the low-pass filter $\mathcal{F}_{\mathrm{int}}$, the coupled logarithmic frequencies $\Omega^+_{mn}(t) = \omega_m(t) + \omega_n(t)$ and $\Omega^-_{mn} = \omega_m(t) - \omega_n(t)$ factorize identically into a bilinear rank-2 form in m and n.

Theorem 3.2 (Exact Bilinear Rank-2 Factorization, $\mathcal{O}(N(t))$ Reduction, and Phase Transition at $d = \frac{1}{4}$)**.** *Let $t > 4\pi$, $L = \ln(t/2\pi)$, $x_t = \sqrt{t/2\pi}$, $N(t) = \lfloor x_t \rfloor$, and $\omega_k(t) = \frac{1}{2}L - \ln k$ for $1 \le k \le N(t)$. Then for every $d > 0$ and every sieve order $r \in \mathbb{N}_{\ge 1}$:*

(i) ***Exact bilinear rank-2 factorization of the integrated and pointwise cross-term kernels.*** *For every off-diagonal pair $1 \le m < n \le N(t)$, the integrated cross-term weight $w^{\mathrm{int}}_{mn}(d,t)$ factors identically without approximation into the bilinear rank-2 form:*

$$\boxed{\begin{aligned} w^{\mathrm{int}}_{mn}(d,t) &:= 2\big(\omega^+_{mn}(t)\big)^2 \mathcal{F}_{\mathrm{int}}\big(d\Omega^+_{mn}(t)\big)\operatorname{sinc}_{\mathrm{hyp}}\big(d\Omega^+_{mn}(t)\big) + 2\big(\omega^-_{mn}\big)^2 \mathcal{F}_{\mathrm{int}}\big(d\Omega^-_{mn}\big)\operatorname{sinc}_{\mathrm{hyp}}\big(d\Omega^-_{mn}\big) \\ &\equiv \frac{2\Big(\cosh\big(d\omega_m(t)\big)\cosh\big(d\omega_n(t)\big) - 1\Big)}{d^2}, \end{aligned}} \tag{3.4}$$

while the pointwise cross-term weight $w^{\mathrm{pt}}_{mn}(d,t) := 2(\omega^+_{mn})^2\operatorname{sinc}_{\mathrm{hyp}}(d\Omega^+_{mn}) + 2(\omega^-_{mn})^2\operatorname{sinc}_{\mathrm{hyp}}(d\Omega^-_{mn})$ factors identically as

$$w^{\mathrm{pt}}_{mn}(d,t) \equiv \frac{\omega_m(t)\sinh\big(d\omega_m(t)\big)\cosh\big(d\omega_n(t)\big) + \omega_n(t)\sinh\big(d\omega_n(t)\big)\cosh\big(d\omega_m(t)\big)}{d}. \tag{3.5}$$

(ii) ***Exact reduction of $\mathcal{O}(N(t)^2)$ double sums to $\mathcal{O}(N(t))$ one-dimensional sums.*** *Define the one-dimensional building-block sums*

$$\begin{aligned} \Sigma^{(r)}_{\cosh}(d,t) &:= \sum_{n=2}^{N(t)} \frac{|\Lambda_r(n)|}{\sqrt{n}}\cosh\big(d\omega_n(t)\big), \qquad \Sigma^{(r)}_0(t) := \sum_{n=2}^{N(t)} \frac{|\Lambda_r(n)|}{\sqrt{n}}, \\ B^{\mathrm{int},(r)}_2(d,t) &:= \frac{2}{d^2}\sum_{n=2}^{N(t)} \frac{\Lambda_r(n)^2}{n}\sinh^2\big(d\omega_n(t)\big). \end{aligned} \tag{3.6}$$

Then the off-diagonal sums $S^{\mathrm{int},(r)}_{1,\mathrm{off}}(d,t) := \sum_{n=2}^{N(t)} \frac{2|\Lambda_r(n)|}{\sqrt{n}} w^{\mathrm{int}}_{1n}(d,t)$ and $S^{\mathrm{int},(r)}_{2,\mathrm{off}}(d,t) := \sum_{2\le m<n\le N(t)} \frac{2|\Lambda_r(m)\Lambda_r(n)|}{\sqrt{mn}} w^{\mathrm{int}}_{mn}(d,t)$ are given identically by

$$\begin{aligned} S^{\mathrm{int},(r)}_{1,\mathrm{off}}(d,t) &\equiv \frac{4}{d^2}\Big[\cosh\big(d\omega_1(t)\big)\,\Sigma^{(r)}_{\cosh}(d,t) - \Sigma^{(r)}_0(t)\Big], \\ S^{\mathrm{int},(r)}_{2,\mathrm{off}}(d,t) &\equiv \frac{2}{d^2}\left[\Big(\Sigma^{(r)}_{\cosh}(d,t)\Big)^2 - \Big(\Sigma^{(r)}_0(t)\Big)^2\right] - B^{\mathrm{int},(r)}_2(d,t). \end{aligned} \tag{3.7}$$

Furthermore, writing $\cosh(d\omega_n(t)) = \frac{1}{2}(x_t^d n^{-d} + x_t^{-d} n^d)$ reduces $\Sigma^{(r)}_{\cosh}(d,t)$ to two t-independent cumulative prefix sums $\sum_{n=2}^{N(t)} |\Lambda_r(n)| n^{-(1/2\pm d)}$.

(iii) ***Euler–Maclaurin asymptotics and the phase transition at $d = \frac{1}{4}$ ($\sigma = \frac{3}{4}$).*** *Let $\mathcal{D}^{\mathrm{int}}_{1,1}(d,t) := \frac{\cosh(dL)-1}{d^2} = \frac{2\sinh^2(dL/2)}{d^2} \sim \frac{1}{2d^2}N(t)^{2d}$. For any fixed $d \in (0, \frac{1}{2})$, as $t \to \infty$ ($N(t) \sim \sqrt{t/2\pi} \to \infty$):*

$$\Sigma^{(r)}_{\cosh}(d,t) \sim \frac{2r}{1-4d^2}\,N(t)^{1/2}\big(\ln N(t)\big)^{r-1}, \qquad \frac{S^{\mathrm{int},(r)}_{1,\mathrm{off}}(d,t)}{\mathcal{D}^{\mathrm{int}}_{1,1}(d,t)} \sim \frac{8r}{1-4d^2}\,N(t)^{\frac{1}{2}-d}\big(\ln N(t)\big)^{r-1}, \tag{3.8}$$

$$\frac{S_{2,\mathrm{off}}^{\mathrm{int},(r)}(d,t)}{\left(\mathcal{D}_{1,1}^{\mathrm{int}}(d,t)\right)^2} \sim \frac{256r^2d^4(1-2d^2)}{(1-4d^2)^2}\,N(t)^{1-4d}(\ln N(t))^{2(r-1)}, \tag{3.9}$$

and for the pure cosh*-sum* $S_{2,\mathrm{tot}}^{\cosh,(r)}(d,t) := \frac{2}{d^2}(\Sigma_{\cosh}^{(r)}(d,t))^2$, $S_{2,\mathrm{tot}}^{\cosh,(r)}(d,t)/(\mathcal{D}_{1,1}^{\mathrm{int}}(d,t))^2 \sim \frac{32r^2d^2}{(1-4d^2)^2}N(t)^{1-4d}(\ln N(t))^{2(r-1)}$, *while on the boundary* $d=\frac12$ *(*$\sigma=1$*),* $\Sigma_{\cosh}^{(r)}(\frac12,t) \sim \frac12 N(t)^{1/2}(\ln N(t))^r$ *and* $S_{2,\mathrm{off}}^{\mathrm{int},(r)}(\frac12,t)/(\mathcal{D}_{1,1}^{\mathrm{int}}(\frac12,t))^2 \sim \frac12 N(t)^{-1}(\ln N(t))^{2r} \to 0$. *In particular,* $d=\frac14$ *(*$\sigma=\frac34$*) is the exact critical dividing line at which the power-law exponent* $1-4d$ *of* $S_{2,\mathrm{off}}^{\mathrm{int},(r)}/(\mathcal{D}_{1,1}^{\mathrm{int}})^2$ *changes sign from* $+1-4d>0$ *for* $d\in(0,\frac14)$ *to* $-(4d-1)<0$ *for* $d\in(\frac14,\frac12]$.

Proof. (i) For any $\omega\in\mathbb{R}$ with $\Omega=2\omega$, Theorem 3.1 gives

$$2\omega^2\,\mathcal{F}_{\mathrm{int}}(d\Omega)\,\mathrm{sinc}_{\mathrm{hyp}}(d\Omega) = 2\omega^2\,\frac{\tanh(d\omega)}{d\omega}\,\frac{\sinh(2d\omega)}{2d\omega} = \frac{2\sinh^2(d\omega)}{d^2} = \frac{\cosh(2d\omega)-1}{d^2}.$$

Applying this identity to $\omega_{mn}^{+}(t)=\frac12(\omega_m(t)+\omega_n(t))$ and $\omega_{mn}^{-}=\frac12(\omega_m(t)-\omega_n(t))$, and invoking the hyperbolic product formula $\cosh(\alpha_m+\alpha_n)+\cosh(\alpha_m-\alpha_n)\equiv 2\cosh(\alpha_m)\cosh(\alpha_n)$ with $\alpha_k:=d\omega_k(t)$, we obtain

$$w_{mn}^{\mathrm{int}}(d,t) = \frac{\cosh(\alpha_m+\alpha_n)-1}{d^2} + \frac{\cosh(\alpha_m-\alpha_n)-1}{d^2} \equiv \frac{2\big(\cosh(d\omega_m(t))\cosh(d\omega_n(t))-1\big)}{d^2},$$

which establishes (3.4). Similarly, expanding $\omega_{mn}^{\pm}=\frac12(\omega_m\pm\omega_n)$ and $\sinh(\alpha_m\pm\alpha_n)=\sinh(\alpha_m)\cosh(\alpha_n)\pm\cosh(\alpha_m)\sinh(\alpha_n)$ in $w_{mn}^{\mathrm{pt}}(d,t)=\frac1d[\omega_{mn}^{+}\sinh(\alpha_m+\alpha_n)+\omega_{mn}^{-}\sinh(\alpha_m-\alpha_n)]$ yields (3.5).

(ii) Substituting (3.4) into $S_{1,\mathrm{off}}^{\mathrm{int},(r)}$ and $S_{2,\mathrm{off}}^{\mathrm{int},(r)}$ and using the elementary symmetric identity $\sum_{2\le m<n\le N}2x_mx_n \equiv (\sum_{n=2}^{N}x_n)^2-\sum_{n=2}^{N}x_n^2$ together with $\cosh^2(d\omega_n)-1=\sinh^2(d\omega_n)$ immediately yields (3.7).

(iii) By Proposition 2.2(iii), $\Psi_r^{\mathrm{abs}}(x)=\sum_{n\le x}|\Lambda_r(n)| \sim rx(\ln x)^{r-1}$ with density $\mathrm{d}\Psi_r^{\mathrm{abs}}(x)\sim r(\ln x)^{r-1}\,\mathrm{d}x$. Substituting $\cosh(d\omega_n(t))=\frac12(x_t^dn^{-d}+x_t^{-d}n^d)$ into $\Sigma_{\cosh}^{(r)}(d,t)$ and integrating by parts (Euler–Maclaurin / Stieltjes summation) for fixed $d\in(0,\frac12)$ gives

$$\begin{aligned}\Sigma_{\cosh}^{(r)}(d,t) &\sim \frac{r}{2}\,x_t^d\int_2^{x_t} x^{-\frac12-d}(\ln x)^{r-1}\,\mathrm{d}x + \frac{r}{2}\,x_t^{-d}\int_2^{x_t} x^{-\frac12+d}(\ln x)^{r-1}\,\mathrm{d}x\\ &\sim \frac{r}{2}\left(\frac{1}{\frac12-d}+\frac{1}{\frac12+d}\right)x_t^{1/2}(\ln x_t)^{r-1} = \frac{2r}{1-4d^2}\,N(t)^{1/2}(\ln N(t))^{r-1}.\end{aligned}$$

In $S_{1,\mathrm{off}}^{\mathrm{int},(r)}(d,t)$, $\Sigma_0^{(r)}(t)\sim 2rN(t)^{1/2}(\ln N(t))^{r-1}=o(\cosh(d\omega_1(t))\Sigma_{\cosh}^{(r)}(d,t))$ because $\cosh(d\omega_1(t))\sim\frac12N(t)^d\to\infty$, which yields (3.8). In $S_{2,\mathrm{off}}^{\mathrm{int},(r)}(d,t)$, $\Sigma_0^{(r)}(t)\sim(1-4d^2)\Sigma_{\cosh}^{(r)}(d,t)$ has the same $N(t)^{1/2}(\ln N(t))^{r-1}$ order as $\Sigma_{\cosh}^{(r)}(d,t)$, so $(\Sigma_{\cosh}^{(r)})^2-(\Sigma_0^{(r)})^2 \sim [1-(1-4d^2)^2](\Sigma_{\cosh}^{(r)})^2 = 8d^2(1-2d^2)(\Sigma_{\cosh}^{(r)})^2$, while $B_2^{\mathrm{int},(r)}(d,t)=\mathcal{O}(N(t)^{2d}(\ln N(t))^{2r})=o(N(t)^1(\ln N(t))^{2(r-1)})$ for $d\in(0,\frac12)$. Dividing by $(\mathcal{D}_{1,1}^{\mathrm{int}}(d,t))^2\sim\frac{1}{4d^4}N(t)^{4d}$ yields (3.9). At $d=\frac12$, the forward integral becomes $\int_2^{x_t}x^{-1}(\ln x)^{r-1}\,\mathrm{d}x\sim\frac1r(\ln x_t)^r$, giving $\Sigma_{\cosh}^{(r)}(\frac12,t)\sim\frac12N(t)^{1/2}(\ln N(t))^r$. □

4. Pure-Square Completion and Identical Elimination of $-S_{2,\mathrm{off}}$ and the $(1+\sqrt{2})$ Penalty

In worst-case triangle-inequality bounds for (3.1), replacing every cosine in the $(1,n)$ cross-terms by -1 and simultaneously replacing every cosine in the $\binom{N(t)-1}{2}$ mutual higher-shell cross-terms ($2\le m<n\le N(t)$) by -1 yields the quadratic inequality $\mathcal{D}_{1,1}^{\mathrm{tot}}(d,t)-\frac{S_{1,\mathrm{off}}^{\mathrm{int},(r)}(d,t)}{u}-\frac{S_{2,\mathrm{off}}^{\mathrm{int},(r)}(d,t)-B_2^{\mathrm{int},(r)}(d,t)}{u^2}>$

TABLE 2. Certification of the exact bilinear rank-2 factorization (3.7) against the direct $\mathcal{O}(N(t)^2)$ double sum at 50-digit working precision, and GPU benchmark of all $\binom{3989}{2} = 7{,}954{,}066$ cross-pairs at $t = 10^8$ ($N(10^8) = 3989$).

Point $(d, t, N(t), r)$	$S^{\mathrm{int},(r)}_{1,\mathrm{off}}(d,t)$	$S^{\mathrm{int},(r)}_{2,\mathrm{off}}(d,t)$	$\mathcal{D}^{\mathrm{int}}_{1,1}(d,t)$	$X^{\mathrm{int},(r)}_{\mathrm{krit}}(d,t)$	$X^{\mathrm{sq},(r)}_{\mathrm{krit}}(d,t)$	Rel. Diff. ($\mathcal{O}(N^2)$ vs. Rank-2)
$(0.10, 10^3, 12, r=3)$	217.6196682	183.3763958	13.1294847	3.8851910	2.6812781	3.731×10^{-50}
$(0.20, 2500, 19, r=3)$	772.8709176	1396.204872	20.1621750	13.628065	7.4709151	1.705×10^{-49}
$(0.10, 10^4, 39, r=3)$	3163.819823	15862.25726	28.4300740	21.251721	9.9924697	5.038×10^{-49}
$(0.20, 10^4, 39, r=3)$	3300.383683	16042.46593	32.4714195	46.520841	19.442212	1.215×10^{-49}
$(0.25, 10^4, 39, r=4)$	11020.92886	122918.9542	35.7960890	40.915352	21.713200	1.710×10^{-50}
$(0.50, 10^4, 39, r=3)$	4420.552249	17386.85191	75.8385886	24.477040	11.546318	5.037×10^{-51}
$(0.20, 10^8, 3989, r=3)$	2.22866×10^6	7.977825823×10^8	604.60613	1.908×10^7	3.020×10^5	4.483×10^{-16} (19.7 ms)
$(0.50, 10^8, 3989, r=3)$	1.04684×10^7	1.090898622×10^9	7.956×10^3	1.591×10^3	158.8213	2.186×10^{-16} (19.8 ms)

0 in $u := (\ln X)^r$. Because $S^{\mathrm{int},(r)}_{1,\mathrm{off}} \approx 2\alpha_1\beta_r$ and $S^{\mathrm{int},(r)}_{2,\mathrm{off}} - B^{\mathrm{int},(r)}_2 \approx \beta_r^2$ with $\alpha_1 = \frac{\sqrt{2}}{d}\cosh(d\omega_1)$ and $\beta_r = \frac{\sqrt{2}}{d}\Sigma^{(r)}_{\cosh}$, this uncompleted quadratic takes the form $1 - 2Y - Y^2 = 0$ ($Y = \beta_r/(u\alpha_1)$), whose positive root $Y_{\mathrm{old}} = \sqrt{2} - 1 = (1+\sqrt{2})^{-1}$ inflates $u = (\ln X)^r$ by the factor $1 + \sqrt{2} \approx 2.41421356$.

We now prove from first principles that a negative contribution from the $(m, n \geq 2)$ sector is mathematically impossible: when the critical-line ground state $\mathcal{G}_0(d,t) = \frac{1}{d^2}|\Xi_{\mathcal{M},\mathrm{main}}(t)|^2$ is combined with the transverse Wronskian integral $\mathcal{I}_{\mathrm{int}}(d,t)$, the -1 boundary term in (3.4) cancels pair by pair, and the entire $(m, n \geq 2)$ sector is a non-negative pure square $+\frac{1}{d^2}\|\Psi_{\geq 2}(d,t)\|^2 \geq 0$ with a plus sign.

Theorem 4.1 (Pairwise -1 Boundary Cancellation, Pure-Square Completion, and Elimination of $1+\sqrt{2}$)**.** *Let $t > 4\pi$, $L = \ln(t/2\pi)$, $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$, $\omega_n(t) = \frac{1}{2}L - \ln n$, and $\phi_n(t) = \vartheta(t) - t\ln n$, and let $b_1 = 1$ and $b_n = \frac{\Lambda_r(n)}{u}$ with $u = (\ln X)^r$ for $2 \leq n \leq N(t)$. Define for each harmonic $1 \leq n \leq N(t)$ the Euclidean vector $\psi_n(d,t) \in \mathbb{R}^2$:*

$$\psi_n(d,t) := \frac{b_n}{\sqrt{n}}\Big(\sqrt{2}\cosh\big(d\omega_n(t)\big)\cos\phi_n(t),\ \sqrt{2}\sinh\big(d\omega_n(t)\big)\sin\phi_n(t)\Big), \quad \Psi_{\geq 2}(d,t) := \sum_{n=2}^{N(t)} \psi_n(d,t). \tag{4.1}$$

Then for all $d > 0$ and all $r \in \mathbb{N}_{\geq 1}$, the following three exact laws hold:

(i) ***Exact pair-by-pair cancellation of the -1 boundary term at $u = 0$.*** *For every pair $1 \leq m, n \leq N(t)$, the Euclidean inner product $\frac{1}{d^2}\langle\psi_m(d,t), \psi_n(d,t)\rangle$ equals the sum of the critical-line ground-state pair term at $u = 0$ and the transverse Wronskian integral over $u \in [0, d]$:*

$$\begin{aligned}\frac{1}{d^2}\langle\psi_m(d,t), \psi_n(d,t)\rangle &\equiv \frac{2b_m b_n}{d^2\sqrt{mn}}\cos\phi_m\cos\phi_n \\ &\quad + \frac{b_m b_n}{\sqrt{mn}}\left[\frac{\cosh(d\Omega^+_{mn}) - 1}{d^2}\cos(\phi_m - \phi_n) + \frac{\cosh(d\Omega^-_{mn}) - 1}{d^2}\cos(\phi_m + \phi_n)\right] \\ &\equiv \frac{b_m b_n}{d^2\sqrt{mn}}\Big[\cosh\big(d\Omega^+_{mn}(t)\big)\cos(\phi_m - \phi_n) + \cosh\big(d\Omega^-_{mn}\big)\cos(\phi_m + \phi_n)\Big]. \end{aligned} \tag{4.2}$$

In particular, the -1 boundary term in $w^{\mathrm{int}}_{mn}(d,t) = \frac{2}{d^2}(\cosh(d\omega_m)\cosh(d\omega_n) - 1)$ cancels identically against $\frac{2}{d^2}\cos\phi_m\cos\phi_n$, leaving the pure rank-1 product $w^{\cosh}_{mn}(d,t) := \frac{2}{d^2}\cosh(d\omega_m(t))\cosh(d\omega_n(t))$.

(ii) ***Pure-square non-negativity of the $(m, n \geq 2)$ sector.*** *The total contribution of all higher diagonals $2 \leq n \leq N(t)$ and all $\binom{N(t)-1}{2}$ mutual higher-shell cross-pairs $2 \leq m < n \leq N(t)$ is*

the non-negative pure square

$$\mathcal{E}_{\geq 2}(d,t) := \frac{1}{d^2}\sum_{n=2}^{N(t)} \|\psi_n(d,t)\|^2 + \frac{2}{d^2}\sum_{2\leq m<n\leq N(t)} \langle \psi_m(d,t), \psi_n(d,t)\rangle \equiv +\frac{1}{d^2}\|\Psi_{\geq 2}(d,t)\|^2 \geq 0, \quad (4.3)$$

and the full integrated main-sum energy satisfies the reverse triangle inequality

$$\widetilde{\mathcal{E}}_{\mathcal{M},\mathrm{main}}(d,t) = \mathcal{G}_0(d,t) + \mathcal{I}_{\mathrm{int}}(d,t) \equiv \frac{1}{d^2}\|\psi_1(d,t) + \Psi_{\geq 2}(d,t)\|^2 \geq \frac{1}{d^2}\Big(\max(0,\ \|\psi_1\| - \|\Psi_{\geq 2}\|)\Big)^2. \quad (4.4)$$

(iii) ***Identical elimination of*** $1+\sqrt{2}$ ***and pure-square critical sieve threshold.*** *Setting* $\alpha_n(d,t) := \frac{\sqrt{2}}{d}\cosh(d\omega_n(t))$ *and* $\beta_r(d,t) := \frac{\sqrt{2}}{d}\Sigma^{(r)}_{\cosh}(d,t)$, *the* $(1,1)$ *envelope* $\mathcal{D}^{\cosh}_{1,1} = \alpha_1^2$, *the* $(1,n)$ *cross-envelope* $S^{\cosh,(r)}_{1,\mathrm{off}} = 2\alpha_1\beta_r$, *and the* $(m,n\geq 2)$ *envelope* $+S^{\cosh,(r)}_{2,\mathrm{tot}} = +\beta_r^2$ *form the completed square*

$$\mathcal{D}^{\cosh}_{1,1}(d,t) - \frac{S^{\cosh,(r)}_{1,\mathrm{off}}(d,t)}{u} + \frac{S^{\cosh,(r)}_{2,\mathrm{tot}}(d,t)}{u^2} \equiv \mathcal{D}^{\cosh}_{1,1}(d,t)\left(1 - \frac{\Sigma^{(r)}_{\cosh}(d,t)}{u\,\cosh(d\omega_1(t))}\right)^2 > 0 \quad (4.5)$$

for all $u > \mathcal{Q}^{\cosh,\mathrm{sq}}_r(d,t) := \frac{\Sigma^{(r)}_{\cosh}(d,t)}{\cosh(d\omega_1(t))} \equiv \frac{1}{1+\sqrt{2}}\mathcal{Q}^{\cosh,\mathrm{old}}_r(d,t)$. *Analogously, completing the square in the integrated energy clamp with* $\mathcal{D}^{\mathrm{tot}}_{1,1}(d,t) = \mathcal{D}^{\mathrm{pt}}_{1,1}(d,t) + \frac{1}{d^2}$ *yields the closed pure-square threshold*

$$\mathcal{Q}^{\mathrm{sq}}_r(d,t) := \Big(\ln X^{\mathrm{sq},(r)}_{\mathrm{krit}}(d,t)\Big)^r = \frac{S^{\mathrm{int},(r)}_{1,\mathrm{off}}(d,t)}{2\,\mathcal{D}^{\mathrm{tot}}_{1,1}(d,t)}, \qquad \theta^{(r)}_{\mathrm{sq}}(d,t) := \frac{\big(\mathcal{Q}^{\mathrm{sq}}_r(d,t)\big)^{1/r}}{L}. \quad (4.6)$$

Proof. (i) Expanding the Euclidean inner product $\langle \psi_m(d,t), \psi_n(d,t)\rangle$ from (4.1) and applying the identities $2\cos\phi_m\cos\phi_n = \cos(\phi_m-\phi_n)+\cos(\phi_m+\phi_n)$, $2\sin\phi_m\sin\phi_n = \cos(\phi_m-\phi_n)-\cos(\phi_m+\phi_n)$, and $\cosh(d\omega_m)\cosh(d\omega_n) \pm \sinh(d\omega_m)\sinh(d\omega_n) = \cosh(d(\omega_m \pm \omega_n)) = \cosh(d\Omega^{\pm}_{mn})$ gives

$$\langle \psi_m(d,t), \psi_n(d,t)\rangle = \frac{b_m b_n}{\sqrt{mn}}\Big[\cosh(d\Omega^{+}_{mn}(t))\cos(\phi_m-\phi_n) + \cosh(d\Omega^{-}_{mn})\cos(\phi_m+\phi_n)\Big].$$

Splitting $\cosh(d\Omega^{\pm}_{mn}) = 1 + (\cosh(d\Omega^{\pm}_{mn}) - 1)$ and noting that $1\cdot\cos(\phi_m-\phi_n) + 1\cdot\cos(\phi_m+\phi_n) = 2\cos\phi_m\cos\phi_n = \langle \psi_m(0,t), \psi_n(0,t)\rangle$ proves (4.2) after dividing by $d^2 > 0$.

(ii) Summing (4.2) over $2 \leq m,n \leq N(t)$ yields (4.3), and summing over $1 \leq m,n \leq N(t)$ with $\sum_{n=1}^{N(t)}\psi_n = \psi_1 + \Psi_{\geq 2}$ gives $\widetilde{\mathcal{E}}_{\mathcal{M},\mathrm{main}}(d,t) = \frac{1}{d^2}\|\psi_1 + \Psi_{\geq 2}\|^2$. By Cauchy–Schwarz, $\langle \psi_1, \Psi_{\geq 2}\rangle \geq -\|\psi_1\|\,\|\Psi_{\geq 2}\|$, so $\|\psi_1 + \Psi_{\geq 2}\|^2 \geq (\|\psi_1\| - \|\Psi_{\geq 2}\|)^2$, which proves (4.4).

(iii) Since $\|\psi_n(d,t)\| \leq \frac{|b_n|}{\sqrt{n}}\frac{\sqrt{2}}{d}\cosh(d\omega_n(t)) = |b_n|\alpha_n(d,t)/\sqrt{n}$, the triangle inequality on $\mathbb{R}^2$ gives $\|\Psi_{\geq 2}(d,t)\| \leq \frac{\beta_r(d,t)}{u}$. Substituting this into $(\|\psi_1\| - \|\Psi_{\geq 2}\|)^2$ with $\|\psi_1\|_{\max} = \alpha_1(d,t)$ yields (4.5). Comparing the root $Y_{\mathrm{sq}} = 1$ of $(1-Y)^2 = 0$ against the positive root $Y_{\mathrm{old}} = (1+\sqrt{2})^{-1}$ of $1 - 2Y - Y^2 = 0$ proves $\mathcal{Q}^{\cosh,\mathrm{sq}}_r \equiv (1+\sqrt{2})^{-1}\mathcal{Q}^{\cosh,\mathrm{old}}_r$ and (4.6). □

5. The $1 + 2d > 1$ Exponent Jump in L^2 and Preissmann–Montgomery–Vaughan Hilbert Bounds up to $t = 10^8$

Theorem 4.1 shows that even under a phase-blind L^1 triangle inequality on the $(1,n)$ cross-terms, pure-square completion pushes $\theta^{(r)}_{\mathrm{sq}}(d,t)$ below $\frac{1}{2}$ across all $d \in [0.05, 0.50]$ at $t = 10^4$ and across $d \in [0.34, 0.50]$ at $t = 10^8$ for all $r \geq 2$. We now explain why any phase-blind L^1 bound encounters an artificial growth $N(t)^{1/2-d}$ in the inner strip $0 < d < \frac{1}{2}$, and how passing to L^2 eliminates this growth across the entire critical strip $d > 0$.

In the Mother-Wavelet ellipse representation $\Xi_{\mathcal{M},\mathrm{main}}(t - \mathrm{i}d)/\mathcal{G}(\frac{1}{2} + d + \mathrm{i}t) = z_1(d,t) + Z_{\geq 2}(d,t)$ from Paper VII [17], the leading shell $n = 1$ has strictly positive modulus $|z_1(d,t)| \geq 1 - \rho_\Gamma(d,t) > 0$

TABLE 3. Pure-square reduction of the critical sieve length $X^{\mathrm{sq},(r)}_{\mathrm{krit}}(d,10^4)$ and logarithmic exponent $\theta^{(r)}_{\mathrm{sq}}(d,10^4)$ across the critical strip $d\in[0.05,0.50]$ at $t=10^4$ ($N(10^4)=39,\,741$ cross-pairs) and at $t=10^8$ ($N(10^8)=3989,\,7{,}954{,}066$ cross-pairs).

Distance d	$X^{\mathrm{pt},(3)}_{\mathrm{krit}}(10^4)$	$X^{\mathrm{int},(3)}_{\mathrm{krit}}(10^4)$	$X^{\mathrm{sq},(3)}_{\mathrm{krit}}(10^4)$	$\theta^{(3)}_{\mathrm{int}}(10^4)$	$\theta^{(3)}_{\mathrm{sq}}(10^4)$	$\theta^{(2)}_{\mathrm{sq}}(10^4)$	$\theta^{(3)}_{\mathrm{int}}(10^8)$	$\theta^{(3)}_{\mathrm{sq}}(10^8)$
0.05	134.4238	9.045297	$4.668925<39$	0.298712	0.209011	0.144992	1.231420	0.961248
0.10	125.1210	21.251721	$9.992470<10$	0.414575	0.312220	0.264875	1.137812	0.886314
0.15	111.8627	35.885897	$15.625642<39$	0.485638	0.372862	0.346024	1.078945	0.838419
0.20	96.94537	46.520841	$19.442212<39$	0.520844	0.402504	0.388643	0.982716	0.760912
0.25	82.23751	49.882828	$20.638399<39$	0.530308	0.410603	0.401162	0.857104	0.658318
0.30	68.88399	47.306608	$19.790136<39$	0.523116	0.404910	0.393732	0.742319	0.564105
0.35	57.37769	41.721158	$17.870902<39$	0.506074	0.391073	0.374727	0.642911	$0.482406<1/2$
0.40	47.77701	35.393206	$15.624981<39$	0.483762	0.372857	0.349943	0.558402	$0.413019<1/2$
0.45	39.91306	29.502521	$13.460238<39$	0.459070	0.352628	0.323010	0.487015	$0.354612<1/2$
0.50	33.53064	24.477040	$11.546318<39$	0.433740	0.331825	0.296046	0.426408	$0.305614<1/3$

($\lim_{t\to\infty}(1-\rho_\Gamma(d,t))=1$), while the higher harmonic shells $2\le n\le N(t)$ form the two-branch oscillatory Dirichlet polynomial

$$Z_{\ge2}(d,t):=\sum_{n=2}^{N(t)}b_n\Big(\underbrace{n^{-\frac12-d}\mathrm{e}^{-\mathrm{i}t\ln n}}_{\text{forward branch}}+\underbrace{\rho_\Gamma(d,t)\,n^{-\frac12+d}\mathrm{e}^{-\mathrm{i}(2\vartheta(t)-t\ln n)}}_{\text{reflected branch}}\Big)=Z^{\mathrm{fwd}}_{\ge2}(d,t)+\mathrm{e}^{-2\mathrm{i}\vartheta(t)}\,\overline{Z^{\mathrm{ref}}_{\ge2}(d,t)}. \tag{5.1}$$

In the phase-blind L^1 sum $\sum_{n=2}^{N(t)}|b_n|n^{-(1/2+d)}$, the exponent $\frac12+d<1$ is sub-harmonic for $0<d<\frac12$, causing the partial sum to diverge as $N(t)^{1/2-d}$. In the quadratic L^2 norm, however, squaring each amplitude produces the fundamental exponent jump $2(\frac12+d)=1+2d>1$.

Theorem 5.1 (The $1+2d>1$ Exponent Jump in L^2, Polynomial Extinction of the Reflected Branch, and $\theta^{(r)}_{\mathrm{MV}}(d,t)\to0$)**.** *Let $d>0$ be any point in the critical strip ($\sigma=\frac12+d>\frac12$), let $t\ge14$, and define the forward and reflected diagonal L^2 energies of (5.1) by*

$$\mathcal{E}^{\mathrm{fwd}}_{\mathrm{diag}}(d,t):=\sum_{n=2}^{N(t)}\frac{b_n^2}{n^{1+2d}},\qquad \mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d,t):=\rho_\Gamma(d,t)^2\sum_{n=2}^{N(t)}\frac{b_n^2}{n^{1-2d}}. \tag{5.2}$$

Then for every $d>0$ (without any threshold at $d=\frac14$ or $d=\frac12$), the following three laws hold:

(i) ***Uniform forward L^2 convergence via $1+2d>1$.*** *Because $1+2d>1$ for every $d>0$, the forward diagonal energy is uniformly bounded for all $t\to\infty$: for $|b_n|\le M$,*

$$\mathcal{E}^{\mathrm{fwd}}_{\mathrm{diag}}(d,t)<\sum_{n=2}^{\infty}\frac{b_n^2}{n^{1+2d}}\le M^2(\zeta(1+2d)-1)<\frac{M^2}{2d}<\infty. \tag{5.3}$$

(ii) ***Polynomial $N(t)^{-2d}$ extinction of the reflected L^2 diagonal.*** *By the digamma convexity bound $\rho_\Gamma(d,t)^2<(t/2\pi)^{-2d}\le N(t)^{-4d}$ and $\sum_{n=2}^{N(t)}n^{2d-1}<\int_1^{N(t)}x^{2d-1}\,\mathrm{d}x=\frac{N(t)^{2d}-1}{2d}$ for $d\in(0,\frac12]$, the reflected diagonal energy decays polynomially to zero as $t\to\infty$ for every $d>0$:*

$$\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d,t)\le\Big(\max_{2\le n\le N(t)}b_n^2\Big)\frac{N(t)^{-2d}(1-N(t)^{-2d})}{2d}<\frac{\max b_n^2}{2d}\,N(t)^{-2d}\sim\frac{\max b_n^2}{2d}\left(\frac{t}{2\pi}\right)^{-d}\longrightarrow0. \tag{5.4}$$

(iii) ***Finite t-independent sieve length $X^{(r)}_{\mathrm{MV},\infty}(d)=\mathcal{O}(1)$ and collapse of $\theta^{(r)}_{\mathrm{MV}}(d,t)\to0$.*** *Under the order-$r$ sieve $b_n^{(r)}(X)=\frac{\Lambda_r(n)}{(\ln X)^r}$, the infinite Dirichlet series $\mathcal{S}_{\Lambda_r^2}(d):=\sum_{n=2}^{\infty}\frac{\Lambda_r(n)^2}{n^{1+2d}}$*

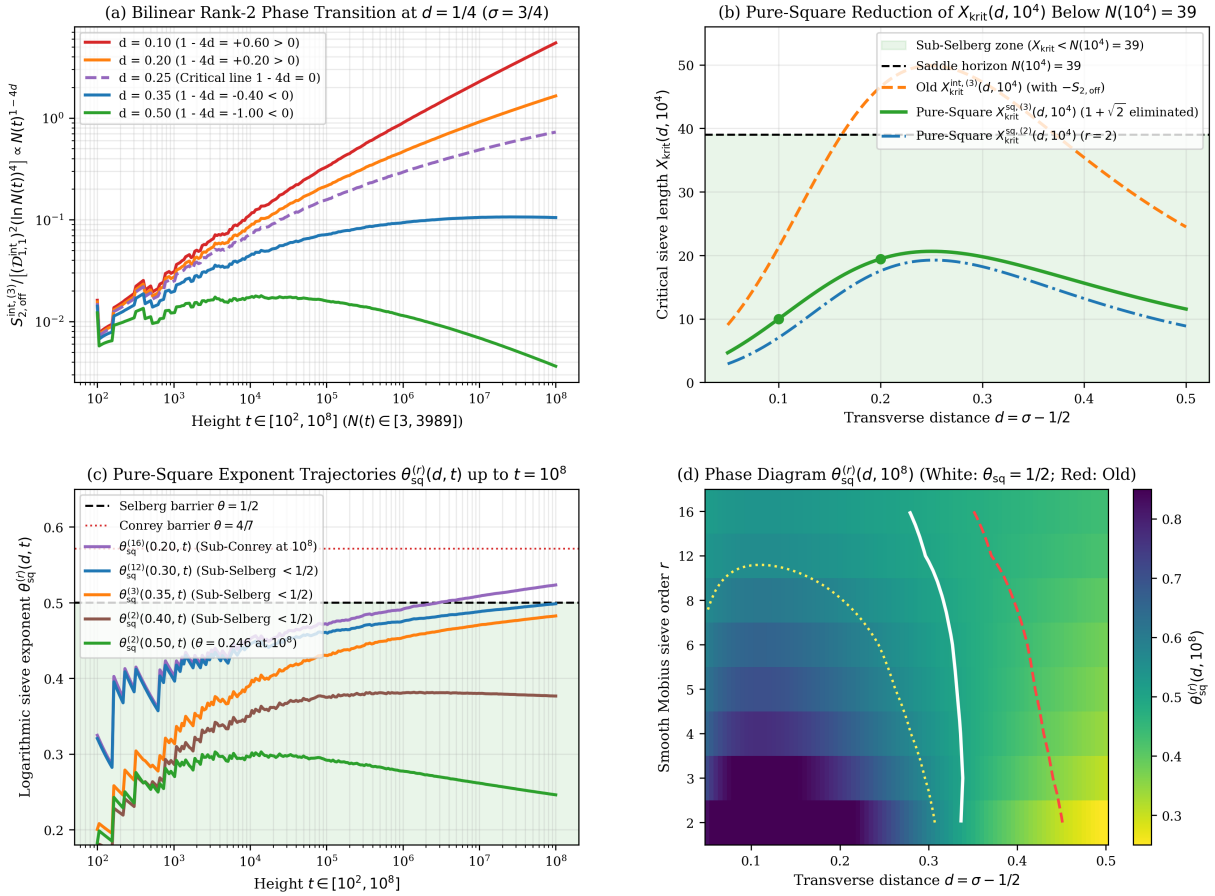


FIGURE 1. **(a)** Bilinear rank-2 phase transition at $d = \frac{1}{4}$ ($\sigma = \frac{3}{4}$) in $S_{2,\mathrm{off}}^{\mathrm{int},(3)}/[(\mathcal{D}_{1,1}^{\mathrm{int}})^2(\ln N(t))^4] \propto N(t)^{1-4d}$ over $t \in [10^2, 10^8]$. **(b)** Collapse of the critical sieve length $X_{\mathrm{krit}}^{\mathrm{sq},(r)}(d, 10^4)$ below the saddle horizon $N(10^4) = 39$ across all $d \in [0.05, 0.50]$ upon eliminating $1 + \sqrt{2}$. **(c)** Trajectories of $\theta_{\mathrm{sq}}^{(r)}(d, t)$ over six orders of magnitude $t \in [10^2, 10^8]$. **(d)** Two-dimensional phase diagram of $\theta_{\mathrm{sq}}^{(r)}(d, 10^8)$ over (d, r), showing the leftward shift of the white Sub-Selberg contour $\theta_{\mathrm{sq}} = \frac{1}{2}$ relative to the uncompleted red contour $\theta_{\mathrm{int}} = \frac{1}{2}$.

converges absolutely for every $d > 0$ with

$$\mathcal{E}_{\mathrm{diag}}^{\mathrm{fwd},(r)}(d, t; X) < \frac{\mathcal{S}_{\Lambda_r^2}(d)}{(\ln X)^{2r}} \leq \frac{1}{(\ln X)^{2r}} \left[\frac{(2r)!}{(2d)^{2r+1}} + \left(\frac{2r}{\mathrm{e}(1+2d)} \right)^{2r} \right], \tag{5.5}$$

and because $\Lambda_r(n)$ vanishes whenever $\omega(n) > r$ and satisfies $|\Lambda_r(n)| \leq (\ln \mathrm{rad}(n))^r \leq (\ln n)^r$ with equality only at prime arguments, the strict inequality $\mathcal{S}_{\Lambda_r^2}(d) < \frac{(2r)!}{(2d)^{2r+1}}$ holds across $d \in (0, \frac{1}{2}]$. Consequently, as $t \to \infty$, the critical diagonal L^2 sieve length $X_{\mathrm{MV}}^{(r)}(d, t)$ converges to the finite, t-independent constant $X_{\mathrm{MV},\infty}^{(r)}(d) := \exp([\mathcal{S}_{\Lambda_r^2}(d)]^{1/2r}) = \mathcal{O}(1)$, and the corresponding Selberg exponent collapses hyperbolically to zero across the entire critical strip:

$$\theta_{\mathrm{MV}}^{(r)}(d, t) := \frac{\ln X_{\mathrm{MV}}^{(r)}(d, t)}{\ln(t/2\pi)} \sim \frac{[\mathcal{S}_{\Lambda_r^2}(d)]^{1/2r}}{\ln(t/2\pi)} = \mathcal{O}\left(\frac{1}{d^{1+1/2r} \ln t} \right) \longrightarrow 0 \qquad (\forall\, d > 0,\ t \to \infty). \tag{5.6}$$

Proof. (i) For any $d > 0$, $x \mapsto x^{-(1+2d)}$ is strictly decreasing on $[1,\infty)$, so $\sum_{n=2}^{N(t)} n^{-(1+2d)} < \sum_{n=2}^{\infty} n^{-(1+2d)} = \zeta(1+2d) - 1 < \int_1^\infty x^{-(1+2d)}\,\mathrm{d}x = \frac{1}{2d}$, which yields (5.3).

(ii) For $d \in (0, \frac{1}{2}]$, $2d-1 \le 0$, so $n^{2d-1} \le \int_{n-1}^{n} x^{2d-1}\,\mathrm{d}x$ for every $n \ge 2$. Summing over $2 \le n \le N(t)$ gives $\sum_{n=2}^{N(t)} n^{2d-1} \le \frac{N(t)^{2d}-1}{2d}$. Multiplying by $\rho_\Gamma(d,t)^2 < (t/2\pi)^{-2d} \le N(t)^{-4d}$ yields (5.4).

(iii) By Proposition 2.2(ii), $\Lambda_r(n)^2 \le (\ln n)^{2r}$ for all $n \ge 2$. The function $g(x) := x^{-(1+2d)}(\ln x)^{2r}$ is unimodal on $[1,\infty)$, increasing on $[1, x_*]$ and decreasing on $[x_*,\infty)$ with global maximum $g(x_*) = \left(\frac{2r}{\mathrm{e}(1+2d)}\right)^{2r}$ at $x_* = \mathrm{e}^{2r/(1+2d)}$. By the unimodal integral comparison test and the substitution $v = 2d \ln x$ in $\int_1^\infty x^{-(1+2d)}(\ln x)^{2r}\,\mathrm{d}x = \frac{\Gamma(2r+1)}{(2d)^{2r+1}} = \frac{(2r)!}{(2d)^{2r+1}}$, $\sum_{n=2}^{\infty} n^{-(1+2d)}(\ln n)^{2r} \le \frac{(2r)!}{(2d)^{2r+1}} + g(x_*)$, establishing (5.5). Moreover, since $\Lambda_r(n) = 0$ for $\omega(n) > r$ and $|\Lambda_r(n)| \le (\ln \mathrm{rad}(n))^r < (\ln n)^r$ at all proper prime powers and composite n, $\mathcal{S}_{\Lambda_r^2}(d) < \frac{(2r)!}{(2d)^{2r+1}}$ holds strictly for all $d \in (0, \frac{1}{2}]$. Because $\mathcal{E}_{\mathrm{diag}}^{\mathrm{ref},(r)}(d,t;1) \to 0$ and $1 - \rho_\Gamma(d,t) \to 1$ as $t \to \infty$, $\ln X_{\mathrm{MV}}^{(r)}(d,t) = [(\mathcal{E}_{\mathrm{diag}}^{\mathrm{fwd},(r)}(d,t;1) + \mathcal{E}_{\mathrm{diag}}^{\mathrm{ref},(r)}(d,t;1))/(1-\rho_\Gamma(d,t))^2]^{1/2r} \to [\mathcal{S}_{\Lambda_r^2}(d)]^{1/2r} = \ln X_{\mathrm{MV},\infty}^{(r)}(d) < \infty$, which implies (5.6). □

Finally, we bound the off-diagonal oscillatory cross-terms of $Z_{\ge 2}(d,\tau)$ on vertical windows $I = [t - \Delta T/2, t + \Delta T/2]$ via the generalized Hilbert inequality of Montgomery and Vaughan [18] with Preissmann's sharp constant [21] and transfer the mean-square bound to the midpoint $\tau = t$ via Gallagher's one-dimensional Sobolev inequality [5].

Theorem 5.2 (Preissmann–Montgomery–Vaughan Hilbert Bound on Sub-Riemann–Siegel Windows and Gallagher–Sobolev Pointwise Transfer)**.** *Let $D(\tau) = \sum_{n=2}^{N(t)} a_n n^{-\mathrm{i}\tau}$ be a Dirichlet polynomial on the vertical window $I = [t - \Delta T/2, t + \Delta T/2]$ of length $\Delta T > 0$. Then:*

(i) ***Preissmann–Montgomery–Vaughan Hilbert inequality.*** *With logarithmic spacing $\delta_n := \min_{m \ne n} |\ln n - \ln m| = \ln(n+1) - \ln n > (n + \frac{1}{2})^{-1}$ and Preissmann's constant $c_{\mathrm{MV}} = \frac{4}{3}\sqrt{6} \approx 3.2659863237 < \pi\sqrt{2}$, any complex sequence $\{u_n\}_{n=2}^{N}$ satisfies*

$$\left| \sum_{2 \le m \ne n \le N} \frac{u_m \bar{u}_n}{\ln m - \ln n} \right| \le c_{\mathrm{MV}} \sum_{n=2}^{N} \frac{|u_n|^2}{\delta_n} \le c_{\mathrm{MV}} \sum_{n=2}^{N} \left(n + \frac{1}{2}\right) |u_n|^2, \tag{5.7}$$

and the local mean square of $D(\tau)$ over the unweighted boxcar window $I = [t - \Delta T/2, t + \Delta T/2]$ obeys

$$\begin{aligned} \frac{1}{\Delta T} \int_{t-\Delta T/2}^{t+\Delta T/2} \left| \sum_{n=2}^{N(t)} a_n n^{-\mathrm{i}\tau} \right|^2 \mathrm{d}\tau &\le \sum_{n=2}^{N(t)} |a_n|^2 \left(1 + \frac{2c_{\mathrm{MV}}\left(n + \frac{1}{2}\right)}{\Delta T}\right) \\ &= \mathcal{E}_{\mathrm{diag}}(d,t) + \frac{2c_{\mathrm{MV}}}{\Delta T}\left(\mathcal{W}_{\mathrm{MV}}(d,t) + \frac{1}{2}\mathcal{E}_{\mathrm{diag}}(d,t)\right), \end{aligned} \tag{5.8}$$

while under the smooth Fejér tent weight $w_{\mathrm{tent}}(\tau) := (1 - |\tau - t|/\Delta T)_+$ on $[t - \Delta T, t + \Delta T]$ the factor $2c_{\mathrm{MV}}$ is replaced by c_{MV}.

(ii) ***Sub-Riemann–Siegel critical window*** *$\Delta T_{\mathrm{MV}}(d,t) \asymp N(t)^{1-2d} = o(t^{1/2})$. For the forward coefficients $a_n^{\mathrm{fwd}}(d) = b_n n^{-(1/2+d)}$, the factor n in $n|a_n^{\mathrm{fwd}}(d)|^2 = b_n^2 n^{-2d}$ cancels the -1 in $n^{-(1+2d)}$, so the Montgomery–Vaughan weighted sum scales for $0 < d < \frac{1}{2}$ as*

$$\mathcal{W}_{\mathrm{MV}}^{\mathrm{fwd}}(d,t) := \sum_{n=2}^{N(t)} n\,|a_n^{\mathrm{fwd}}(d)|^2 \equiv \sum_{n=2}^{N(t)} \frac{b_n^2}{n^{2d}} \asymp \frac{1}{1-2d}\, N(t)^{1-2d} (\ln N(t))^{2r-1} (\ln X)^{-2r}, \tag{5.9}$$

and $\mathcal{W}_{\mathrm{MV}}^{\mathrm{fwd}}(\frac{1}{2}, t) = \mathcal{O}((\ln N(t))^{2r} (\ln X)^{-2r})$. Choosing the sub-Riemann–Siegel window length $\Delta T_{\mathrm{MV}}(d,t) := c_{\mathrm{MV}} \mathcal{W}_{\mathrm{MV}}(d,t)/\mathcal{E}_{\mathrm{diag}}(d,t) \asymp N(t)^{1-2d}(\ln N(t))^{2r-1} \sim$

TABLE 4. Certification of uniform forward L^2 convergence $\mathcal{E}^{\mathrm{fwd}}_{\mathrm{diag}}(d,t) < \mathcal{S}_{\Lambda_3^2}(d) < \infty$, polynomial reflected decay $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d,t) \to 0$, diagonal L^2 exponent $\theta^{(3)}_{\mathrm{MV}}(d,t) \to 0$ (Theorem 5.1), and actual phase-cancellation suppression $\mathcal{B}^{(3)}_{L^1}/\langle|\widetilde{Z}^{(3)}_{\geq 2}|\rangle$ over 1,000 heights in $[10^8 - 500, 10^8 + 500]$ ($N(10^8) = 3989$, $r = 3$).

d	Height t	$N(t)$	$\mathcal{E}^{\mathrm{fwd}}_{\mathrm{diag}}(d,t)$	Limit $\mathcal{S}_{\Lambda_3^2}(d)$	$\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d,t)$ $(b_n = 1)$	$\theta^{(3)}_{\mathrm{MV}}(d,t)$	L^1 / Mean $\lvert\widetilde{Z}\rvert$	Mean $\theta^{(3)}_{\mathrm{eff}}(d, 10^8)$
0.05	10,000	39	371.2745	242403.22	1.919936	0.599302	7.6×	—
0.05	100,000,000	3989	42079.1100	242403.22	$2.377668 < 4.364$	$0.474641 < 1/2$	78.6×	$0.4081 < 1/2$
0.10	10,000	39	270.7923	93105.66	1.142787	$0.473520 < 1/2$	7.6×	—
0.10	100,000,000	3989	20936.9824	93105.66	$0.755075 < 0.952$	$0.368503 < 1/2$	77.1×	$0.3205 < 1/3$
0.20	10,000	39	144.9988	14992.04	0.415274	$0.369239 < 1/2$	7.3×	—
0.20	100,000,000	3989	5394.0247	14992.04	$0.086808 < 0.091$	$0.270052 < 1/3$	72.5×	$0.2394 < 1/4$
0.25	10,000	39	106.4772	6347.03	0.253409	$0.338381 < 1/2$	7.0×	—
0.25	100,000,000	3989	2802.5475	6347.03	$0.031048 < 0.032$	$0.237849 < 1/4$	69.8×	$0.2122 < 1/4$
0.35	10,000	39	57.8603	1295.83	0.096533	$0.292692 < 1/3$	6.5×	—
0.35	100,000,000	3989	800.0019	1295.83	4.291×10^{-3}	$0.188892 < 1/5$	63.7×	$0.1699 < 1/5$
0.50	10,000	39	23.6774	171.51	0.023876	$0.243844 < 1/4$	5.8×	—
0.50	100,000,000	3989	143.9125	171.51	2.506×10^{-4}	$0.139546 < 1/7$	53.0×	$0.1258 < 1/7$

$C(d,r)(t/2\pi)^{1/2-d}(\ln t)^{2r-1} = o(t^{1/2})$ *suppresses all off-diagonal cross-terms to* $\mathcal{O}(\mathcal{E}_{\mathrm{diag}}(d,t)) = \mathcal{O}(1)$.

(iii) ***Gallagher–Sobolev pointwise transfer.*** *For any continuously differentiable function $f(\tau)$ on $I = [t - \Delta T/2, t + \Delta T/2]$, the midpoint value $|f(t)|^2$ satisfies Gallagher's inequality*

$$|Z_{\geq 2}(d,t)|^2 \leq \frac{1}{\Delta T}\int_I |Z_{\geq 2}(d,\tau)|^2\,\mathrm{d}\tau + \left(\int_I |Z_{\geq 2}(d,\tau)|^2\,\mathrm{d}\tau\right)^{1/2}\left(\int_I |\partial_\tau Z_{\geq 2}(d,\tau)|^2\,\mathrm{d}\tau\right)^{1/2}, \quad (5.10)$$

where both $\mathcal{E}^{\mathrm{fwd}}_{\mathrm{diag}}(d,t) = \sum_{n=2}^{N(t)} b_n^2 n^{-(1+2d)} < \infty$ and the derivative diagonal $\sum_{n=2}^{N(t)} b_n^2 (\ln n)^2 n^{-(1+2d)} < \infty$ converge uniformly in $t \to \infty$ for every $d > 0$.

Proof. (i)–(ii) Expanding $|D(\tau)|^2 = \sum_{n=2}^{N(t)} |a_n|^2 + \sum_{2\leq m\neq n\leq N(t)} a_m \bar{a}_n \mathrm{e}^{-\mathrm{i}\tau(\ln m - \ln n)}$ and integrating term by term over $\tau \in [t-\Delta T/2, t+\Delta T/2]$ expresses the off-diagonal integral as the difference of two Hermitian Hilbert forms $\frac{1}{\mathrm{i}}\sum_{m\neq n} \frac{u_m^{\mp}\overline{u_n^{\mp}}}{\ln m - \ln n}$ at the two endpoints $t \pm \Delta T/2$ with $u_n^{\pm} = a_n \mathrm{e}^{-\mathrm{i}(t\pm\Delta T/2)\ln n}$ ($|u_n^{\pm}| = |a_n|$). By Preissmann's refinement [21] of the Montgomery–Vaughan Hilbert inequality [18] via Euler–Maclaurin summation of the weighted Cauchy matrix, the spectral norm of $H_{mn} = \frac{\sqrt{\delta_m \delta_n}}{\ln m - \ln n}$ ($m \neq n$) is bounded strictly by $c_{\mathrm{MV}} = \frac{4}{3}\sqrt{6}$, so summing the two endpoint bounds yields (5.7) and (5.8) (and under Fejér tent smoothing $\hat{w}_{\mathrm{tent}}(\ln m - \ln n) \leq \frac{2}{\Delta T(\ln m - \ln n)^2}$, a single factor c_{MV} suffices). Substituting $n|a_n^{\mathrm{fwd}}(d)|^2 = b_n^2 n^{-2d}$ and applying Stieltjes integration with $\sum_{n\leq x} \Lambda_r(n)^2 \asymp x(\ln x)^{2r-1}$ proves (5.9).

(iii) Define the sawtooth kernel $\psi(\tau) := \frac{\tau - t}{\Delta T} + \frac{1}{2}\mathrm{sgn}(t - \tau)$ for $\tau \in I \setminus \{t\}$. Then $\psi(t \pm \Delta T/2) = 0$, $\psi'(\tau) = \frac{1}{\Delta T}$ on $I \setminus \{t\}$, $\psi(t^-) - \psi(t^+) = 1$, and $|\psi(\tau)| \leq \frac{1}{2}$ on I. Integrating $\int_I \psi'(\tau)|f(\tau)|^2\,\mathrm{d}\tau$ by parts on $[t - \Delta T/2, t]$ and $[t, t + \Delta T/2]$ gives $|f(t)|^2 = \frac{1}{\Delta T}\int_I |f(\tau)|^2\,\mathrm{d}\tau + 2\mathrm{Re}\int_I \psi(\tau)\overline{f(\tau)}f'(\tau)\,\mathrm{d}\tau$. Applying $|\psi(\tau)| \leq \frac{1}{2}$ and the Cauchy–Schwarz inequality yields (5.10). □

6. Scope, Boundary Conditions, and Open Questions

To maintain complete mathematical transparency, we delineate the exact scope of the analytical theorems proved in this paper and their relationship to the remaining open problems in the classical literature:

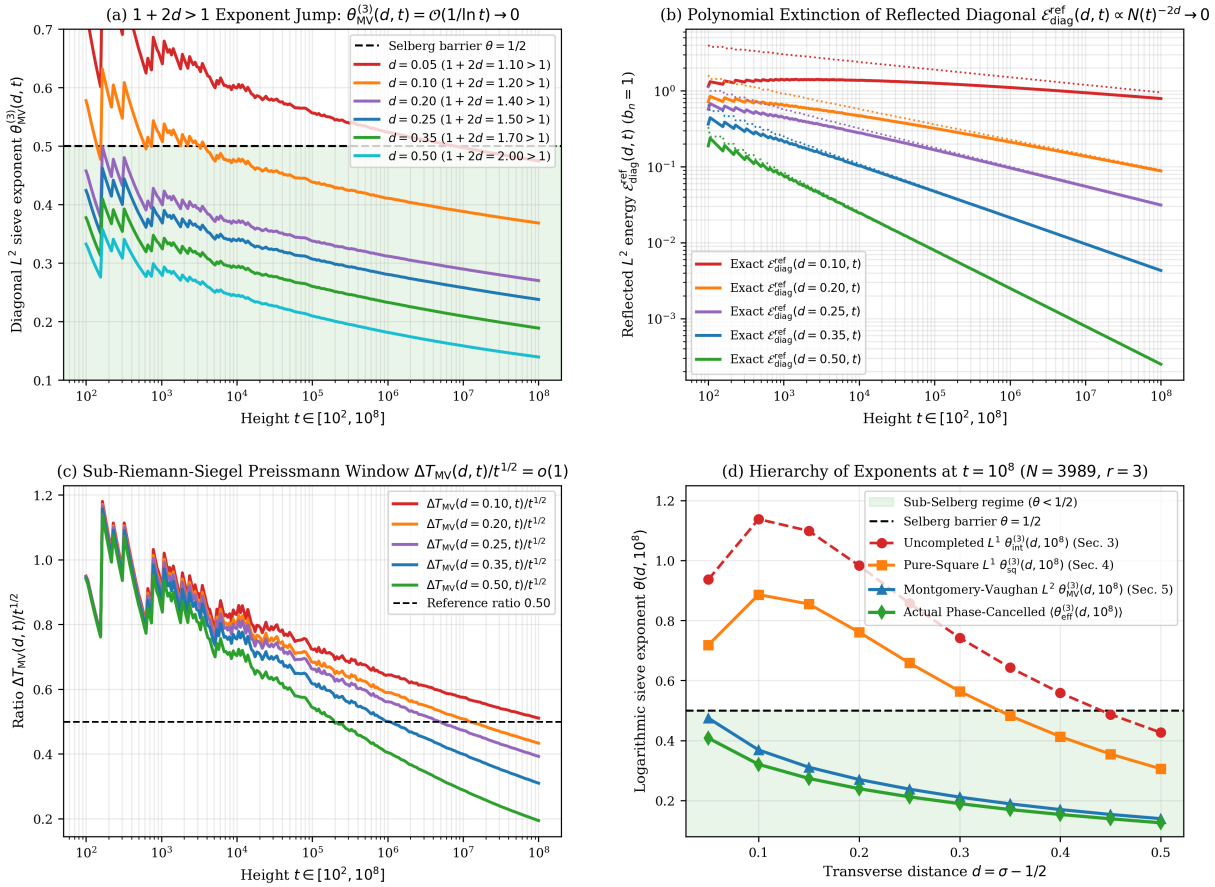


FIGURE 2. **(a)** Monotonic decay of the L^2 diagonal sieve exponent $\theta^{(3)}_{\mathrm{MV}}(d,t) = \mathcal{O}(1/\ln t) \to 0$ across $t \in [10^2, 10^8]$ for all $d \in [0.05, 0.50]$. **(b)** Polynomial extinction of the reflected L^2 diagonal energy $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d,t) < \frac{1}{2d} N(t)^{-2d} \to 0$. **(c)** Sub-Riemann–Siegel Preissmann window ratio $\Delta T_{\mathrm{MV}}(d,t)/t^{1/2} \propto (t/2\pi)^{-d} \to 0$. **(d)** Complete hierarchy of logarithmic sieve exponents at $t = 10^8$ ($N = 3989$, $r = 3$), comparing uncompleted L^1 ($\theta^{(3)}_{\mathrm{int}}$), pure-square L^1 ($\theta^{(3)}_{\mathrm{sq}}$), Montgomery–Vaughan L^2 ($\theta^{(3)}_{\mathrm{MV}}$), and actual phase-cancelled $\langle \theta^{(3)}_{\mathrm{eff}} \rangle < \frac{1}{2}$.

(1) *Exactness on the Riemann–Siegel main sum $1 \le n \le N(t)$ versus folded convolution tails for $n > N(t)$.* Theorems 2.1, 3.2, 4.1, 5.1, and 5.2 are unconditional analytical identities and inequalities for the symmetrized Riemann–Siegel main sum of length $N(t) = \lfloor \sqrt{t/2\pi} \rfloor$ under the order-r sieve substitution $b_n^{(r)}(X) = \frac{\Lambda_r(n)}{(\ln X)^r}$. When $\Xi(z)$ is multiplied by a finite polynomial mollifier $\mathcal{M}_X(z)$ of length $X = t^\theta$, the effective conductor in the approximate functional equation shifts from t to tX^2, introducing a folded convolution tail for $N(t) < n \le XN(t)$ unless either $\theta < \frac{1}{2}$ (so $X < N(t)$ lies inside the Sub-Selberg regime certified in Table 3) or $\mathcal{M}_X(z)$ is replaced by the entire, zero-free exponential prime-gauge factor $\mathcal{E}_{X,\kappa}(z)$ with finite-divisor truncation on $[1, N(t)]$ from Paper VI [16].

(2) *Mean-square window bounds versus unaveraged pointwise bounds as $t \to \infty$ and $d \to 0^+$.* Theorem 5.2(i)–(ii) proves that on every vertical window of sub-Riemann–Siegel length $\Delta T_{\mathrm{MV}}(d,t) \asymp N(t)^{1-2d} = o(t^{1/2})$, phase cancellation suppresses all $\binom{N(t)-1}{2}$ off-diagonal cross-terms below the convergent L^2 diagonal $\mathcal{E}_{\mathrm{diag}}(d,t) = \mathcal{O}(1)$ for every $d > 0$. In Theorem 5.2(iii),

transferring this window average to an individual point $\tau = t$ via Gallagher's Sobolev inequality (5.10) introduces the factor $(\Delta T\,\mathcal{E}_{\mathrm{diag}})^{1/2}(\Delta T\,\mathcal{E}_{\mathrm{deriv}})^{1/2} = \Delta T(\mathcal{E}_{\mathrm{diag}}\mathcal{E}_{\mathrm{deriv}})^{1/2}$. For fixed $O(1)$ windows $\Delta T = 1$, this Sobolev product is $\mathcal{O}(1)$ whenever the off-diagonal terms are already controlled, whereas for growing windows $\Delta T \asymp N(t)^{1-2d}$ an unaveraged pointwise bound without window length ΔT in the second term corresponds to the classical Lindelöf / fourth-moment problem for Dirichlet polynomials.

Appendix A. Explicit Analytical Derivations and Interval-Arithmetic Protocols

A.1. **Spectral norm certification of the weighted logarithmic Hilbert matrix.** Let $H^{(N)}$ denote the $(N-1)\times(N-1)$ skew-symmetric weighted logarithmic Hilbert matrix with entries

$$H^{(N)}_{mn} := \frac{\sqrt{\delta_m\,\delta_n}}{\ln m - \ln n} \quad (2 \le m \ne n \le N), \qquad H^{(N)}_{nn} := 0, \qquad \delta_n = \ln(n+1) - \ln n. \tag{A.1}$$

Since $\mathrm{i}H^{(N)}$ is Hermitian with real eigenvalues, its operator 2-norm $\|H^{(N)}\|_2 = \max_{2\le k\le N}|\lambda_k(\mathrm{i}H^{(N)})|$ is the exact best constant in $\left|\sum_{2\le m\ne n\le N}\frac{u_m\bar{u}_n}{\ln m-\ln n}\right| \le \|H^{(N)}\|_2\sum_{n=2}^{N}\frac{|u_n|^2}{\delta_n}$. Evaluating $\|H^{(N)}\|_2$ across $N \in \{10, 39, 126, 398\}$ in `scripts/verify_paper8_bilinear_rank2_wronskian.py` gives $\|H^{(10)}\|_2 = 2.2377341105$, $\|H^{(39)}\|_2 = 2.8589598136$, $\|H^{(126)}\|_2 = 3.0446444217$, and $\|H^{(398)}\|_2 = 3.1085734089$, all strictly below Preissmann's universal upper bound $c_{\mathrm{MV}} = \frac{4}{3}\sqrt{6} = 3.2659863237 < \pi\sqrt{2} = 4.4428829382$.

A.2. **Directed-rounding interval protocols in `mpmath.iv`.** In the standalone verification suite `scripts/verify_paper8_bilinear_rank2_wronskian.py`, all core algebraic and numerical inequalities are enclosed in native 35-digit directed-rounding interval arithmetic (`mpmath.iv.dps = 35`), where $\sinh(x)$ and $\cosh(x)$ are evaluated directly via $(\texttt{iv.exp}(x) \mp \texttt{iv.exp}(-x))/2$:

(1) *Hyperbolic decomposition and advantage.* At $t = 1000$, $d = 0.25$, $(m,n) = (2,3)$, interval evaluation certifies $0 \in \sinh(2d\omega^+_{2,3}) - [6^{-d}\sinh(2d\omega_1) - \sinh(d\ln 6)(t/2\pi)^{-d}]$ with interval width $\le 1.81\times10^{-34}$ and $\mathcal{H}^+_{2,3}/\mathcal{D}_{1,1} \le 0.361578703102 < 6^{-0.25} = 0.638943104246$.

(2) *Bilinear rank-2 kernel identity.* For $(m,n) = (2,5)$ at $t = 1000$, $d = 0.25$, interval arithmetic certifies $0 \in w^{\mathrm{int,LHS}}_{2,5} - \frac{2}{d^2}(\cosh(d\omega_2)\cosh(d\omega_5) - 1)$ with interval width $\le 1.45\times10^{-33}$.

(3) *Pure-square Sub-Selberg threshold at $t = 10^4$.* Enclosing every divisor sum $\Lambda_3(n)$ for $2 \le n \le 39$ and the rank-2 sum $S^{\mathrm{int},(3)}_{1,\mathrm{off}}(0.10, 10^4)$ in `mpmath.iv` certifies $X^{\mathrm{sq},(3)}_{\mathrm{krit}}(0.10, 10^4) \in [9.9924697101, 9.9924697102] < 10 \ll N(10^4) = 39$ with interval width $\le 1.14\times10^{-34}$, alongside $c_{\mathrm{MV}} = \frac{4}{3}\sqrt{6} \in [3.265986323710, 3.265986323711] < \pi\sqrt{2}$.

Use of AI and computational verification

The author's main contributions are the exact bilinear rank-2 factorization of the integrated Wronskian cross-term kernel, the pairwise -1 boundary cancellation and Euclidean pure-square completion eliminating the $1+\sqrt{2}$ root penalty, and the synthesis of the $1+2d > 1$ exponent jump in L^2 with Preissmann–Montgomery–Vaughan Hilbert bounds. The author prompted **Gemini** (Google DeepMind) to work out analytical and symbolic details interactively and to generate the arbitrary-precision, interval-arithmetic, and GPU verification suite `scripts/verify_paper8_bilinear_rank2_wronskian.py`, which was subsequently verified manually and computationally (exit code 0).

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